A-5 – Maintenance

This section describes maintenance procedures and guidelines for the FastBack® Model 90E. The following topics are discussed in this section:

- Cleaning
- Recommended Maintenance Intervals
- Removing the Pan Assembly
- Replacing the Pan Assembly
- Removing the Enclosure
- Replacing the Enclosure
- Greasing the Connecting Rod Bearings
- Retensioning the Motor Belt
- Replacing the Motor Belt
- Replacing the Motor Assembly
- Retensioning the Eccentric Timing Belt
- Replacing the Eccentric Timing Belt
- Replacing the Crankshaft Assembly

Cleaning

Observe the following guidelines when cleaning the FastBack® Model 90E.

Multiple voltage sources may be present, including control wiring from the scales, PLCs, or other devices. Use the same safety precautions as with any equipment powered by electricity.

Secure all power at the source before applying any water.

WARNING

Limit high-pressure water to product handling areas of the conveyor. Never direct high-pressure water at the photoeyes, drive assembly enclosure, power module, or motor junction box.

CAUTION

Do not use abrasives, scouring powders, or any product that may scratch the equipment surface.

Recommended Maintenance Intervals

Table 5-1 lists recommended maintenance intervals for the FastBack® Model 90E and the type of maintenance to be performed at each interval.

Interval	Recommended Maintenance
Following each cleaning	• Check photoeye calibration • Check pan attachment clamps
Following each shift or pan replacement	• Check pan attachment clamps • Check photoeye calibration
Weekly	• Check tightness of enclosure latches • Check tightness of bolts on articulating feet • Check tightness of locknuts on threaded stems of articulating feet • Check pan attachment clamps
Every 6 months	• Check all belts and retension or replace
Every 12 months	• Grease the connecting rod bearings using a NLGI Grade 2, Polyurea, EP-rated grease with high adhesion

Table 5-1 – Recommended Maintenance Intervals

Removing the Pan Assembly

This procedure describes how to remove the pan from the drive assembly enclosure. Always remove the pan before removing the enclosure.

CAUTION

Always use caution when picking up or moving any heavy object. If you cannot lift 50 pounds or have not removed a pan before, consider having someone assist you.

Step	**Description**
1	**Disconnect power to the FastBack®**

Disconnect power to the FastBack® by rotating the emergency disconnect switch to the OFF position.

WARNING

This step is necessary to prevent the FastBack® from accidentally starting during this procedure.

2 Find pan's center of gravity

Find the pan's center of gravity relative to the four band clamps attaching the pan to the drive assembly. The center of gravity on most Model 90E pans is outside the four clamps, as shown in Figure 5-1.

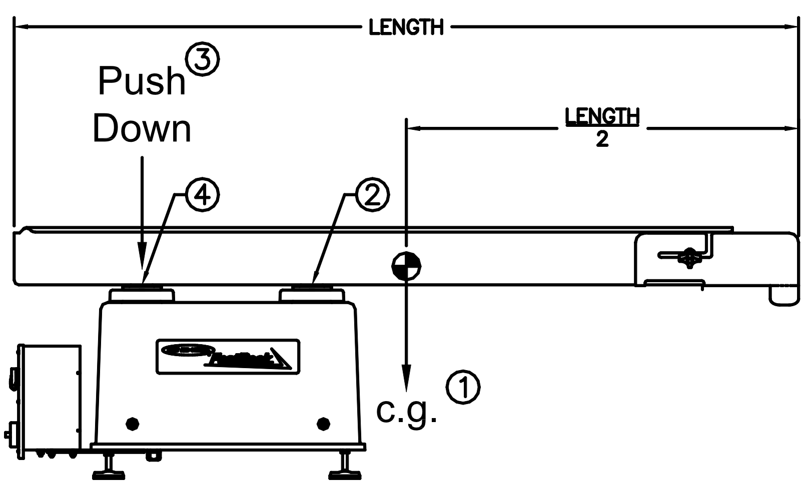

Figure 5-1 - Center of Gravity for Pan Assembly

Step	Description
3	**Unlatch first two clamps**

Unlatch the two clamps nearest to the pan's center of gravity.

CAUTION

Unlatching the two clamps nearest the pan's center of gravity will prevent the pan from tipping nose down with only one or two clamps still attached. If the pan tips, the attachment disks may be damaged or the weld attaching the top disk to the pan may be broken. This step is especially important if you are working alone.

4 Unlatch remaining clamps

Place your free hand in the bottom of the pan over the last two clamps and push downward. This will prevent the pan's nose from tipping down when the clamps are unlatched.

Unlatch the last two clamps, maintaining pressure on the pan with your hand (see Photo 5-1).

Photo 5-1 - Supporting Pan While Unlatching

Step	**Description**
5	**Remove pan**

Remove the pan at its midpoint as shown in Photo 5-2.

CAUTION

You may need to maneuver the pan around other equipment such as other pans or photoeyes. Take care not to bump the pan against a photoeye, which could damage the photoeye or change its calibration.

Photo 5-2 - Lifting the Pan

Place the pan out of the way of foot traffic.

Note: ***To prevent damage to the pan attachment disks, place the pan upside down on the floor as shown in Photo 5-3.***

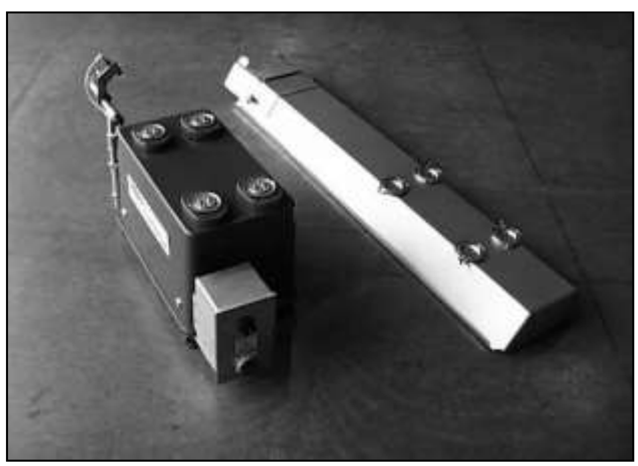

Photo 5-3 - Storing the Pan Upside Down

Replacing the Pan Assembly

This procedure describes how to reattach the pan to the drive assembly.

CAUTION

Always use caution when picking up or moving any heavy object. If you cannot lift 50 pounds by yourself or have not replaced a pan before, consider having someone assist you.

Step	Description
1	**Disconnect power to the FastBack®**

Disconnect power to the FastBack® by rotating the emergency disconnect switch to the OFF position.

WARNING

This step is necessary to prevent the FastBack® from accidentally starting during this procedure.

2	**Replace pan attachment disks if applicable**

Replace the pan attachment disks if they were removed when the pan was removed. Use the disk alignment fixture (part no. 30562434) to align the disks (see Photo 5-4).

Photo 5-4 - Using the Disk Alignment Fixture

Step **Description**

3 Replace band clamps on attachment disks

If the band clamps on the attachment disks were removed, replace them, making sure that they are oriented with the latch on the outer edge of the pan (see Photo 5-5). Maneuver the tabs on the clamp under the edge of the disk, as shown in Photo 5-5.

Photo 5-5 - Replacing the Band Clamps

4 Replace pan

Lifting the pan at its midpoint, reposition it over the drive assembly as shown in Photo 5-6.

Photo 5-6 - Replacing the Pan

Step	Description
5	**Latch band clamps**

Place one hand in the bottom of the pan between the two pairs of band clamps and push downward. This will prevent the pan's nose from tipping down.

Latch the two band clamps nearest the rear of the pan, maintaining pressure on the pan with your hand (see Photos 5-7 and 5-8). Make sure that the band clamp completely captures the top and bottom attachment disks.

Photo 5-7 - Supporting Pan While Latching

Photo 5-8 - Latching the Band Clamp

Latch the remaining two band clamps.

Step	**Description**
5	**Latch band clamps (continued)**

CAUTION

If the band clamp latches insecurely or is difficult to latch, tighten or loosen the adjustment screw using a Phillips screwdriver and an 11/32-inch wrench (see Figure 5-2). Do not operate the FastBack® until the pan is securely attached to the enclosure.

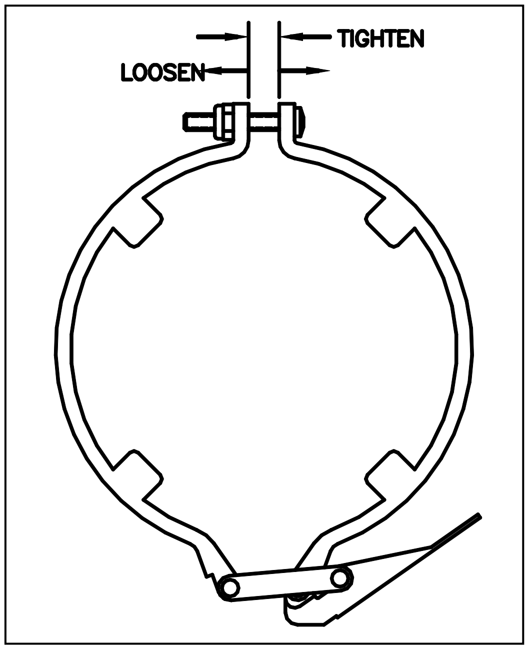

Figure 5-2 – Adjusting the Band Clamp

Removing the Enclosure

This procedure describes how to remove the drive assembly enclosure. Always remove the enclosure completely before performing any maintenance on the drive assembly.

Tools Needed: 7/16-inch or 10-mm socket or wrench, large flat head screwdriver

<u>**Step**</u>	<u>**Description**</u>

1 **Disconnect power to the FastBack®**

Disconnect power to the FastBack® by rotating the emergency disconnect switch to the OFF position.

WARNING ***This step is necessary to prevent the FastBack® from accidentally starting during this procedure.***

2 **Remove pan assembly**

Remove the pan assembly as described in "Removing the Pan Assembly."

3 **Remove bottom pan attachment disks**

Remove the two bolts and lock washer from each bottom pan attachment disk using either a 7/16-inch wrench (for ¼-20 bolts) or a 10-mm wrench (for 6-mm bolts) as shown in Photo 5-9.

Photo 5-9 - Removing Attachment Point Bolts

Step	**Description**
4	**Remove slider covers**

Lift off the bottom pan attachment disk and slider cover (see Photo 5-10).

Photo 5-10 - Removing the Attachment Disk and Slider Cover

If your Model 90E is equipped with slider seal assemblies, remove the brown slider plate, including the four springs on top of the slider (see Photo 5-11).

Note: The springs are not attached to the slider plate.

Photo 5-11 - Removing the Slider Plate

Step	**Description**
5	**Remove enclosure**

Turn the four quarter-turn latches counterclockwise one-quarter turn to release the enclosure (see Photo 5-12).

Note: There is no hard stop at one-quarter turn.

Photo 5-12 – Quarter-Turn Fastener in Unlatched Position

Carefully lift the enclosure straight up and set it aside (see Photo 5-13).

Photo 5-13 - Lifting the Enclosure

Note: You may need to remove additional pans on the surrounding equipment to completely remove the enclosure. Do not perform maintenance on the drive assembly unless the enclosure is completely removed.

Step **Description**

6 Inspect enclosure seal and latch

Inspect the enclosure seal and hook latch to make sure that they are in good condition (see Photo 5-14).

Photo 5-14 - Enclosure Seal and Hook Latch

Replacing the Enclosure

This procedure describes how to replace the drive assembly enclosure.

Tools Needed: 7/16-inch or 10-mm wrench, large flat head screwdriver, torque wrench set to 110 inch-pounds (7/16-inch hardware) or 90 inch-pounds (10-mm hardware)

Step **Description**

1 Disconnect power to the FastBack®

Disconnect power to the FastBack® by rotating the emergency disconnect switch to the OFF position.

WARNING

This step is necessary to prevent the FastBack® from accidentally starting during this procedure.

2 Replace enclosure

Carefully lower the enclosure over the drive assembly (see Photo 5-15). Make sure that the hook cams at each corner are inside the enclosure. To prevent possible damage to the seal on the inside of the enclosure, make sure that nothing contacts the seal as you lower the enclosure.

Photo 5-15 - Replacing the Enclosure

Step	**Description**
2	**Replace enclosure (continued)**

Latch each quarter-turn fastener by rotating it clockwise one-quarter turn. Photo 5-16 shows the fastener in the latched position.

Photo 5-16 - Fastener in Latched Position

3 **Replace slider seal assemblies (if applicable)**

If your Model 90E is equipped with slider seal assemblies, inspect the white slider bases for excessive wear (thickness less than .094 inch) as shown in Figure 5-3.

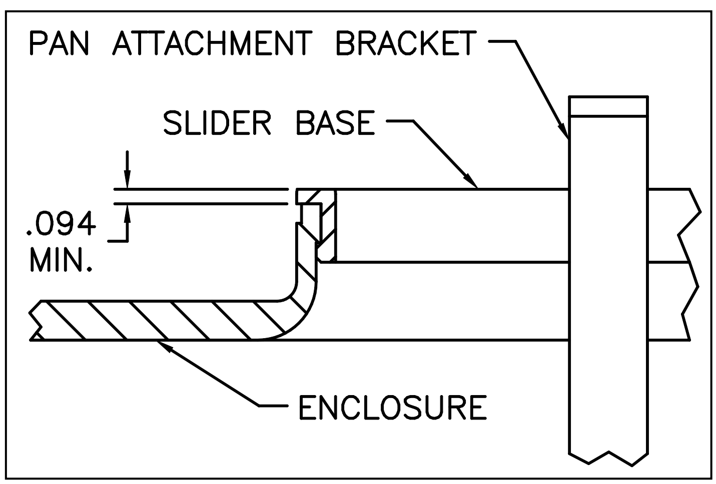

Figure 5-3 – Worn Slider Base

Step	**Description**
3	**Replace slider seal assemblies (if applicable) (continued)**

Replace any worn slider bases with new ones as shown in Photo 5-17.

Photo 5-17 - Replacing the Slider Base

Replace the brown slider plate, including the four springs (see Photo 5-18).

Note: The springs are not attached to the slider plate.

Photo 5-18 - Replacing the Slider Plate

Step	Description

4 **Replace slider covers and pan attachment disks**

Replace the slider covers and the bottom pan attachment disks (see Photo 5-19).

Note: The top and bottom clamp disks must be oriented with the tabs to the outside, as shown in Photo 5-19.

Photo 5-19 - Replacing the Slider Cover and Attachment Disk

Replace the two attachment point bolts using either a 7/16-inch wrench (for ¼-20 bolts) or a 10-mm wrench (for 6-mm bolts), as shown in Photo 5-20. Torque to 110 inch pounds (for ¼-20 bolts) or 90 inch-pounds (for 6-mm bolts).

Photo 5-20 - Replacing the Attachment Point Bolts

5 **Replace pan assembly**

Replace the pan assembly as described in “Replacing the Pan Assembly.”

Greasing the Connecting Rod Bearings

This procedure describes how to grease the Connecting Rod Bearings, which is recommended for annual maintenance.

Step	Description
1	**Disconnect power to the FastBack®**
	Disconnect power to the FastBack® by rotating the emergency disconnect switch to the OFF position.
2	**Remove pan assembly**
	Remove the pan assembly as described in “Removing the Pan Assembly.”
3	**Remove drive assembly enclosure**
	Remove the drive assembly enclosure as described in “Removing the Enclosure.”
4	**Grease the Connecting Rod Bearings**
	Using the Zerk fittings (highlighted in Photo 5-21 below) slowly grease the connecting rod bearings with a NLGI Grade 2, Polyurea, EP-rated grease with high adhesion such as Mobil Polyrex™ EP2. Take care not to blow out the seals which have very small grease vents. Stop adding grease at the first sign of any grease purge.

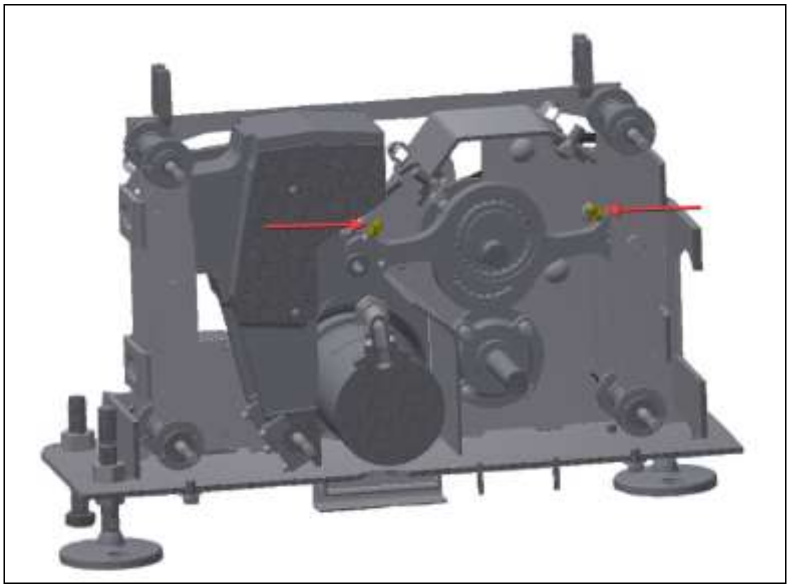

Photo 5-21 – Zerk fitting locations on the 90E Connecting Rods

Retensioning the Motor Belt

This procedure describes how to retension the motor belt.

Tools Needed: 9/16-inch deep socket, 9/16-inch socket, small flat head screwdriver, 4-6-inch extension, torque wrench set to 20 foot-pounds, ruler

<u>**Step**</u> <u>**Description**</u>

5 Disconnect power to the FastBack®

Disconnect power to the FastBack® by rotating the emergency disconnect switch to the OFF position.

6 Remove pan assembly

Remove the pan assembly as described in "Removing the Pan Assembly."

7 Remove drive assembly enclosure

Remove the drive assembly enclosure as described in "Removing the Enclosure."

4 Loosen motor plate nuts

Using a 9/16-inch deep socket and extension, loosen the three motor plate nuts holding the motor plate to the chassis (see Photo 5-21). One-half turn should be sufficient. You should be able to slide the flat washer under each nut by hand.

Photo 5-22 - Loosening the Motor Plate Nuts

Step **Description**

5 **Set belt tension**

Using a 9/16-inch socket and extension, set the belt tension by turning the tensioning bolt until the spring is at a length of 1-1/8 inch +1/32 inch/-0 inch, as shown in Photo 5-22).

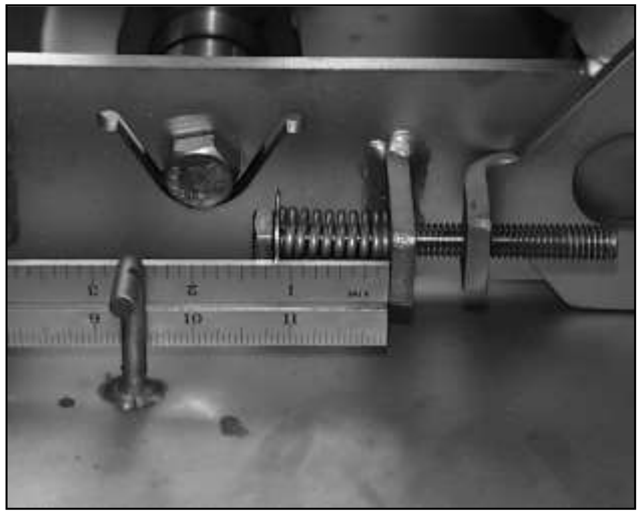

Photo 5-23 - Compression Spring at 1-1/8"

Rotate the mechanism in the normal operating direction by hand several revolutions. This allows the belt to find its natural tracking center.

6 **Verify belt tension**

Using a screwdriver under the bottom edge of the motor plate, move the motor up and down to make sure that it moves freely and that the spring is applying full tension to the belt (see Photo 5-23).

Photo 5-24 - Moving the Motor Plate to Verify Belt Tension

Step **Description**

7 **Tighten motor plate nuts**

Using the 9/16-inch socket and the torque wrench, tighten the three motor plate nuts to 20 foot-pounds (see Photo 5-24).

Photo 5-25 - Tightening the Motor Plate Nuts

8 **Replace enclosure and pan assembly**

Replace the enclosure and pan assembly as described in “Replacing the Enclosure” and “Replacing the Pan Assembly.”

Replacing the Motor Belt

This procedure describes how to replace the motor belt.

Tools Needed: 9/16-inch deep socket, 9/16-inch socket, small flat head screwdriver, 4-6-inch extension, torque wrench set to 20 foot-pounds, ruler

Step	**Description**
1	**Disconnect power to the FastBack®**
	Disconnect power to the FastBack® by rotating the emergency disconnect switch to the OFF position.
2	**Remove pan assembly**
	Remove the pan assembly as described in "Removing the Pan Assembly."
3	**Remove drive assembly enclosure**
	Remove the drive assembly enclosure as described in "Removing the Enclosure."
4	**Loosen tensioning bolt**
	Using a 9/16-inch socket and extension, back the tensioning bolt out far enough to remove any compression in the spring and provide an additional ¼ inch of slack (see Photo 5-25).

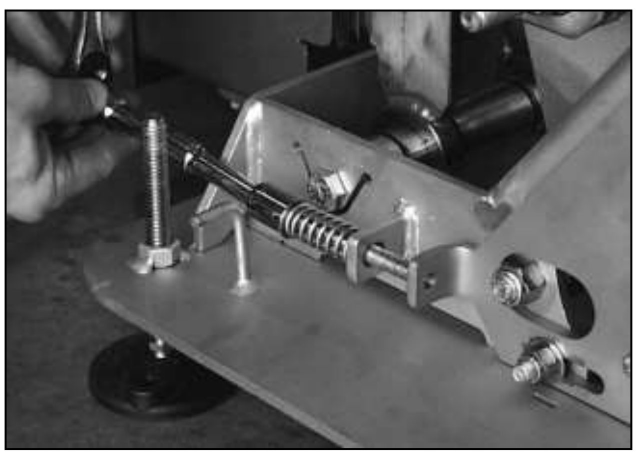

Photo 5-26 - Loosening the Tensioning Bolt

<u>**Step**</u> <u>**Description**</u>

5 Loosen motor plate nuts

Using a 9/16-inch deep socket and extension, loosen the three motor plate nuts holding the motor plate to the chassis (see Photo 5-26). One-half turn should be sufficient. You should be able to slide the flat washer under each nut by hand.

Photo 5-27 - Loosening the Motor Plate Nuts

6 Remove old belt and inspect motor sprocket

Remove the old belt from the motor. Inspect the motor sprocket for damaged teeth or worn plating. If the sprocket is damaged or worn, call the Heat and Control Galesburg, IL office at (309) 342-5518 to request a new sprocket. Grind off the small welds attaching the sprocket to the motor shaft and remove the sprocket using a gear puller.

WARNING

Do not proceed with belt replacement until the new sprocket is installed.

Step	**Description**
7	**Install new belt**

Install the new motor belt by first slipping it over the motor sprocket and then sliding it onto the drive sprocket.

CAUTION

Interference between the motor and the chassis side plate may cause initial belt tension. If this is the case, remove a small amount of material from the chassis at the interference point. Do not go to Step 8 until the interference is removed.

8	**Set belt tension**

Using a 9/16-inch socket and extension, set the belt tension by turning the tensioning bolt until the spring is at a length of 1-1/8 inch +1/32 inch/-0 inch, as shown in Photo 5-27).

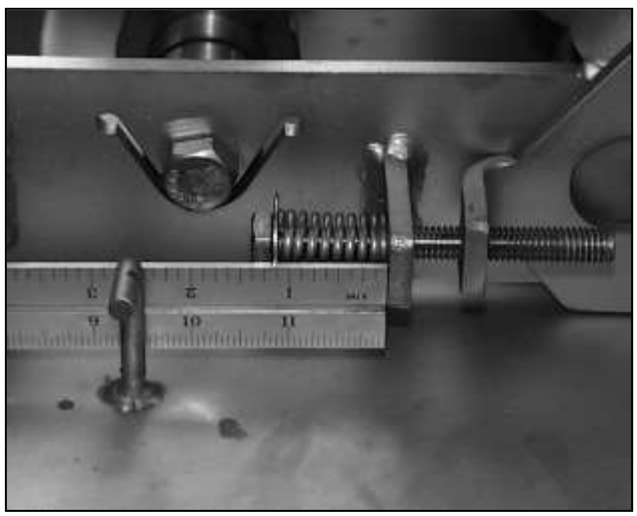

Photo 5-28 - Compression Spring at 1-1/8"

Rotate the mechanism in the normal operating direction by hand several revolutions. This allows the belt to find its natural tracking center.

Step **Description**

9 Verify belt tension

Using a screwdriver under the bottom edge of the motor plate, move the motor up and down to make sure that it moves freely and that the spring is applying full tension to the belt (see Photo 5-28).

Photo 5-29 - Moving the Motor Plate to Verify Belt Tension

10 Tighten motor plate nuts

Using the 9/16-inch socket and the torque wrench, tighten the three motor plate nuts to 20 foot-pounds (see Photo 5-29).

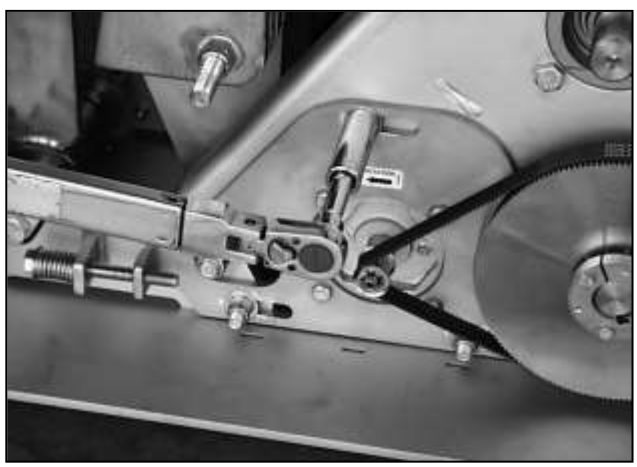

Photo 5-30 - Tightening the Motor Plate Nuts

11 Replace enclosure and pan assembly

Replace the enclosure and pan assembly as described in “Replacing the Enclosure” and “Replacing the Pan Assembly.”

Replacing the Motor Assembly

This procedure describes how to replace a motor and sprocket assembly (the replacement motor includes a sprocket that is welded to the motor). You do not need to remove the old sprocket from the old motor.

Tools Needed: 9/16-inch deep socket, 9/16-inch socket, 7/16-inch socket, small flat head screwdriver, 4- to-6-inch extension, narrow (.100-inch) flat head screwdriver, #2 Phillips screwdriver, torque wrench set to 80 inch-pounds and 20 foot-pounds, ruler

Step	**Description**
1	**Loosen motor plate nuts**

Using a 9/16-inch deep socket and extension, loosen the three motor plate nuts holding the motor plate to the chassis (see Photo 5-30). One-half turn should be sufficient. You should be able to slide the flat washer under each nut by hand.

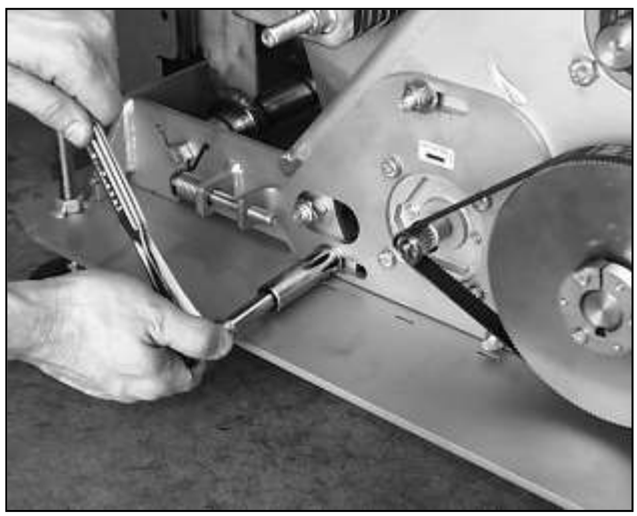

Photo 5-31 - Loosening the Motor Plate Nuts

Step	Description

2 **Remove tension from belt**

Using a 9/16-inch socket and extension, back the tensioning bolt out far enough to remove any compression in the spring and provide an additional ¼ inch of slack (see Photo 5-31).

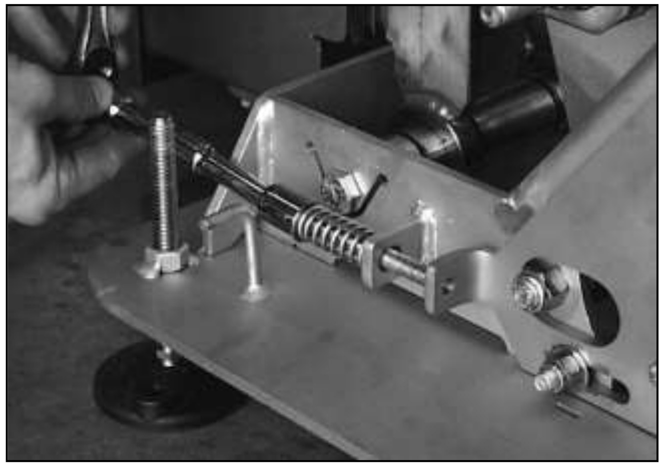

Photo 5-32 - Loosening the Tensioning Bolt

3 **Remove motor belt**

Remove the motor belt by first slipping it off the large drive sprocket. Inspect the motor belt for wear. Discard any belt with signs of uneven wear or fraying.

4 **Remove bolts from motor plate**

Using a 7/16-inch wrench, remove the four bolts and lock washers attaching the motor to the motor plate (see Photo 5-32).

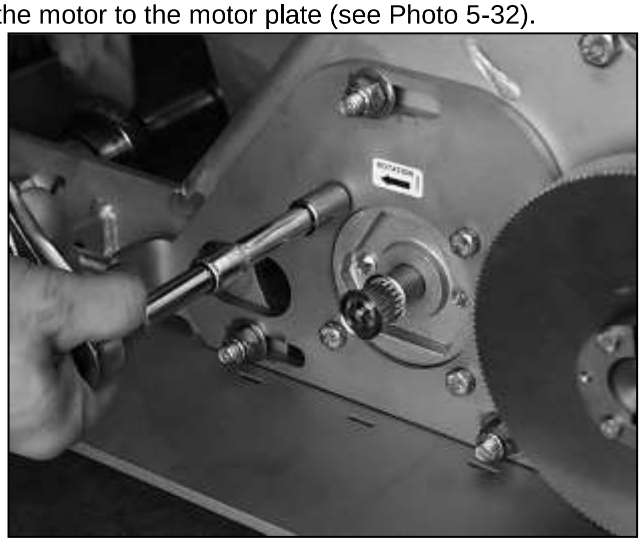

Photo 5-33 - Removing the Motor Plate Bolts

Step	Description
5	**Open motor junction box**

Using the #2 Phillips screwdriver, open the motor junction box on the opposite side of the chassis by loosening the four captive screws on the box lid.

6	**Disconnect motor leads**

Disconnect the motor leads from the terminal block as follows (see Photo 5-33:

1. Insert a narrow screwdriver into the operating slot up to the stop.
2. Hold the screwdriver down to open the clamping spring.
3. Remove the lead.
4. Repeat steps 1 through 3 for each lead in the terminal block.

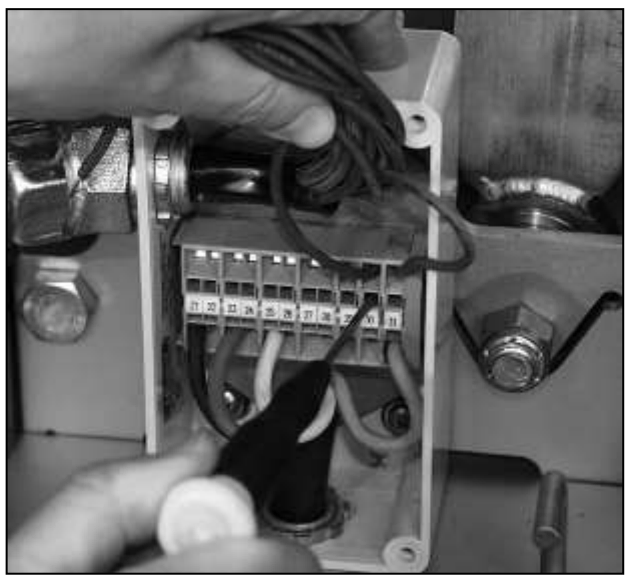

Photo 5-34 - Motor Leads Pulled Through Motor Junction Box

Step **Description**

7 **Remove leads from motor junction box**

Loosen the top nut on the cord grip. Do not apply excessive torque to the top nut, as this may damage the box.

Pull the leads through the cord grip and out of the box (see Photo 5-34).

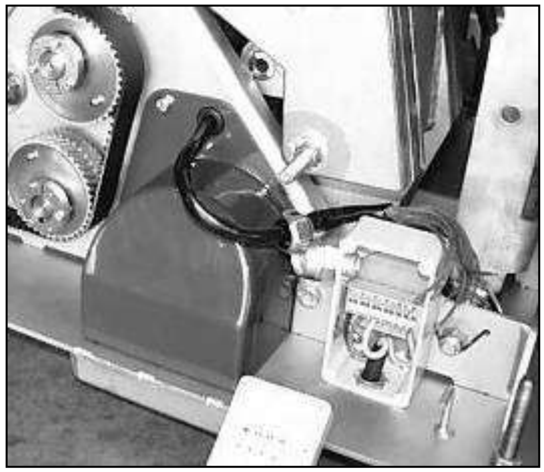

Photo 5-35 - Motor Leads Pulled Through Motor Junction Box

8 **Remove motor shroud**

Rotate the eccentric sprocket system by hand so that the timing belt does not interfere with the left flange of the motor shroud.

Using a 7/16-inch socket, remove the three mounting bolts and lock washers from the motor shroud (see Photo 5-36).

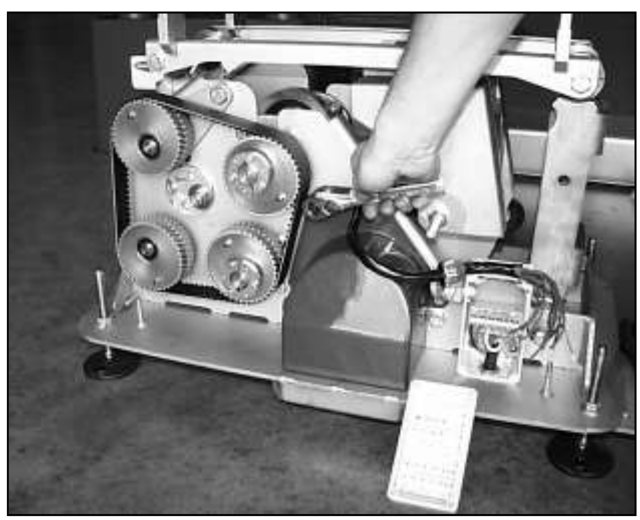

Photo 5-36 - Removing Motor Shroud Nuts

Step	**Description**
8	**Remove motor shroud (continued)**

Pull the motor shroud off completely, taking care to pull the motor leads through the rubber grommet.

9 Remove motor

Slide the motor out through the opening in the chassis plate.

10 Insert new motor

Slide the new motor in through the opening in the chassis plate. The cord grip must be pointing up as shown in Photo 5-37.

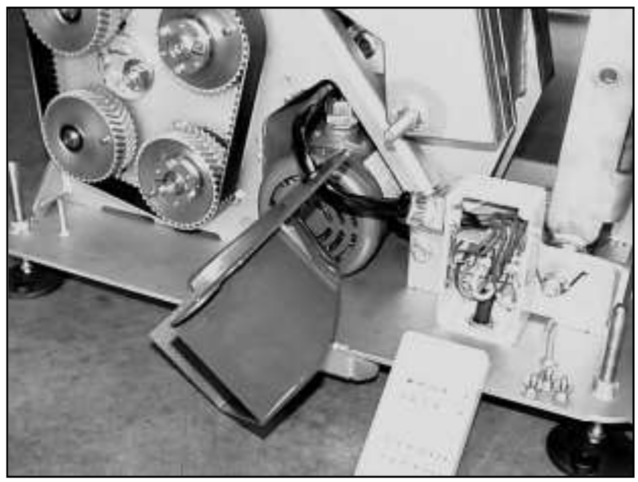

Photo 5-37 - Proper Orientation of New Motor

11 Reattach motor plate

Using the 7/16-inch socket, reattach the four bolts holding the motor to the motor plate. Torque these bolts to 80 inch-pounds.

12 Rewire motor

Run the motor leads through the rubber grommet in the motor shroud and then through the cord grip top nut and grommet of the motor junction box. Trim the new leads to the length of the old motor leads.

Note: The shrink wrap on the leads must be completely inside the cord grip (showing inside the motor junction box).

Strip the insulation back about ¼ inch and tin the tips to prevent fraying.

Step **Description**

13 Wire motor leads to terminal strip

Using the wiring diagram inside the motor junction box lid, attach the motor leads to the terminal strip as follows:

1. Twist the wires at the end of the lead if not tinned.
2. Insert a narrow screwdriver into the operating slot up to the stop.
3. Hold the screwdriver down to open the clamping spring.
4. Insert the lead.
5. Repeat for each lead in the terminal block.

14 Verify motor direction

Verify that the motor turns in the correct direction following the steps below:

WARNING

Do not perform this step with the motor belt in place. The motor will turn at full speed, and the drive mechanism is not balanced with the pan removed.

1. Rotate the emergency disconnect switch to the ON position.
2. Start the motor either by bumping the motor in manual using the PLC or by temporarily jumpering the appropriate terminals on the inverter inside the power module.

   - For Semipower power modules, jumper terminal 8 to terminal 9.
   - For Yaskawa power modules, jumper S1 to SC.

3. Verify the motor direction. The motor shaft should turn counterclockwise as you look at the motor sprocket, as shown on the motor plate sticker.

   If the motor is not turning in the correct direction, switch any two inverter leads (T1 and T2) coming from the power module to the terminal strip inside the motor junction box.

4. Rotate the emergency disconnect switch to the OFF position and replace the lid on the motor junction box, taking care not to pinch any wires in the seal. If your FastBack® is equipped with a Semipower inverter, leave the lid off until you have commissioned the motor (see Step 22)

15 Replace motor shroud

Replace the motor shroud, tightening the three sets of bolts.

Step **Description**

16 Replace motor junction box lid

Replace the four screws on the motor junction box lid.

17 Replace motor belt

Holding the motor plate in place by hand, replace the motor belt by first slipping it over the motor sprocket and then sliding it onto the drive sprocket.

18 Tighten tensioning bolt

Using a 9/16-inch socket and extension, set the belt tension by turning the tensioning bolt until the compression spring is at a length of 1-1/8 inch +1/32 inch/-0 inch, as shown in Photo 5-38.

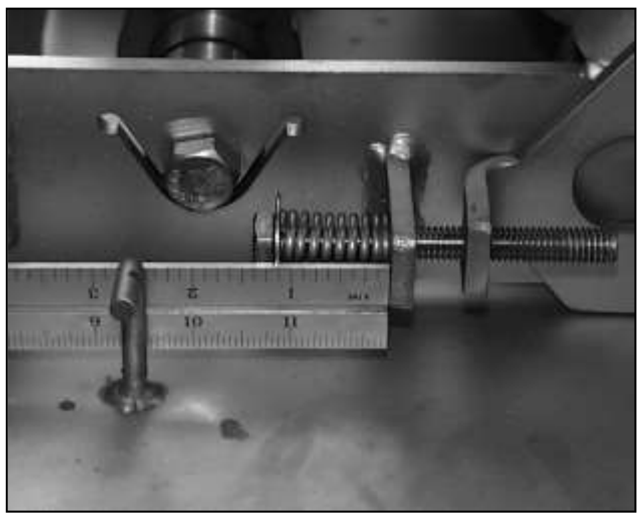

Photo 5-38 - Compression Spring at 1-1/8”

Rotate the mechanism in the normal operating direction by hand several revolutions. This allows the belt to find its natural tracking center.

19 Verify belt tension

Using a screwdriver under the bottom edge of the motor plate, move the motor up and down to make sure that it moves freely and that the spring is applying full tension to the belt (see Photo 5-39).

Photo 5-39 - Moving the Motor to Verify Belt Tension

20 Tighten motor plate nuts

Using a 9/16-inch socket and torque wrench, tighten the three motor plate nuts to 20 foot-pounds (see Photo 5-40).

Photo 5-40 - Tightening the Motor Plate Nuts

Step **Description**

21 **Replace enclosure and pan assembly**

Replace the enclosure and pan assembly as described in “Replacing the Enclosure” and “Replacing the Pan Assembly.”

22 **Commission Semipower inverter if applicable**

If your FastBack® is equipped with a Semipower inverter, Heat and Control strongly recommends recommissioning the inverter for the new motor-inverter combination. Recommissioning ensures that the equipment operates at its highest efficiency.

Recommission the inverter as follows:

1. Turn on power to the FastBack®.

2. Open the power module lid.

3. Temporarily close terminals 8 to 12 to begin the commissioning sequence. The conveyor mechanism may move during commissioning, which takes about 30 seconds.

4. Replace the power module lid.

Retensioning the Eccentric Timing Belt

This procedure describes how to retension the eccentric timing belt of the eccentric sprocket system.

Tools Needed: ¾-inch deep well socket and wrench, ¾-inch socket and extension, 15/16-inch socket and wrench, torque wrench that can be set to 42 and 103 foot-pounds, small machinist's square

<u>**Step**</u>	<u>**Description**</u>
1	**Remove pan and enclosure**
	Remove the pan and enclosure assemblies as described in "Removing the Pan Assembly" and "Removing the Enclosure."
2	**Loosen pan attachment arms**
	Using the ¾-inch wrench and socket, loosen the two bolts on each end of the pan attachment arms. Remove the bolt that goes through the drive arm, shown in Photo 5-41.

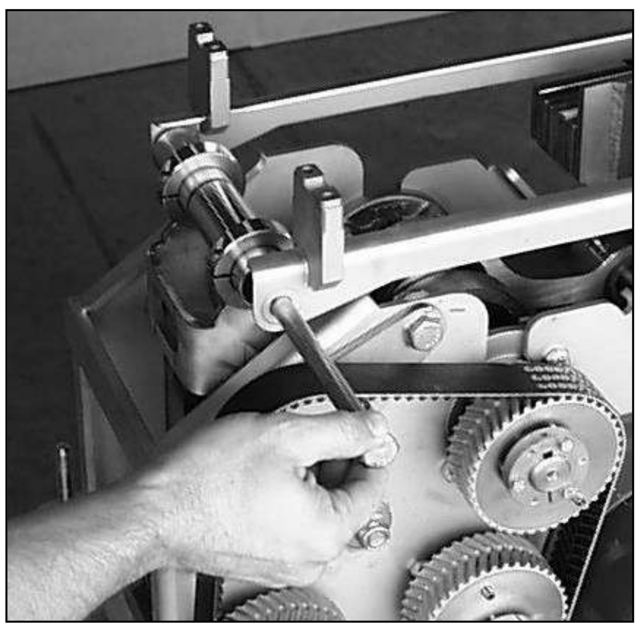

Photo 5-41 - Loosening the Pan Attachment Arm Bolts

Step **Description**

3 Disconnect drive arm from drive arm connecting rod

Using a ¾-inch socket and extension through the chassis access hole, remove the bolt from the drive arm that holds the crank arm in place (see Photo 5-42).

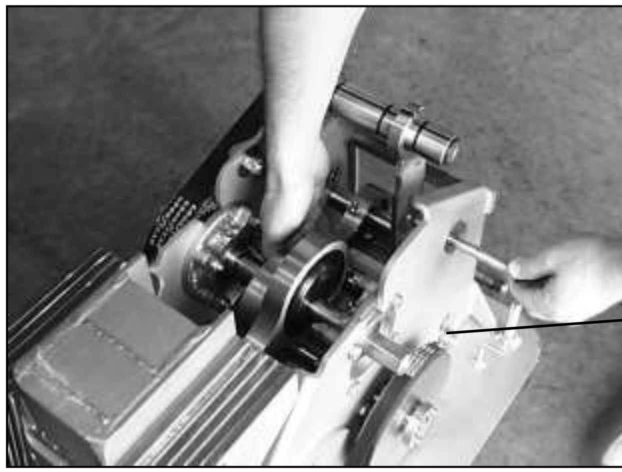

Photo 5-42 - Removing the Drive Arm Bolt

CAUTION

Support the connecting rod with your free hand to prevent it from dropping and damaging the seals in the bearing.

Using the ¾-inch socket and wrench, loosen the lower drive arm bushing bolt (see Photo 5-41, above) so that the drive arm swings freely. Lay the drive arm against the chassis.

Step **Description**

4 Loosen tensioner sprocket pivot nut

Using the ¾-inch wrench, loosen the tensioner sprocket pivot nut one-quarter turn (see Photo 5-43). The nut may be on the opposite side of the frame from that shown.

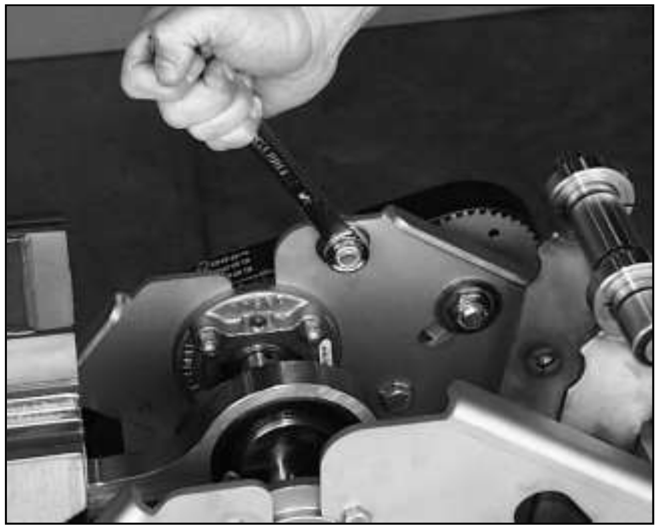

Photo 5-43 - Loosening the Tensioner Sprocket Pivot Nut

5 Loosen jam nut behind tensioner sprocket stud

Using the 15/16-inch socket, loosen the jam nut at the back of the tensioner sprocket stud by about one-half turn (see Photo 5-44).

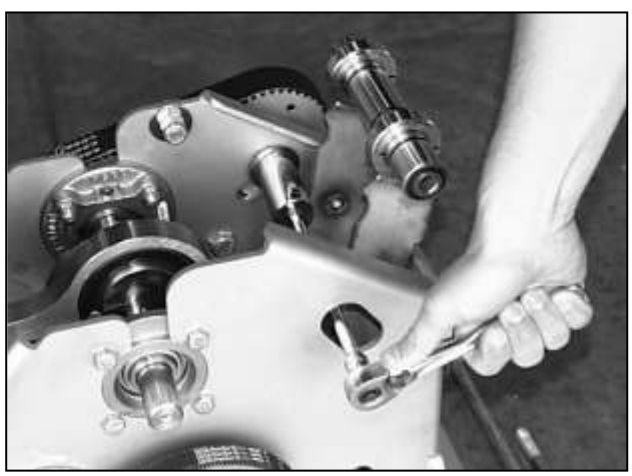

Photo 5-44 - Loosening the Tensioner Sprocket Stud

Step	Description

6 **Loosen nut on tensioner cam pivot bolt**

Using the ¾-inch socket, slightly loosen the nut on the tensioner cam pivot bolt (see Photo 5-45).

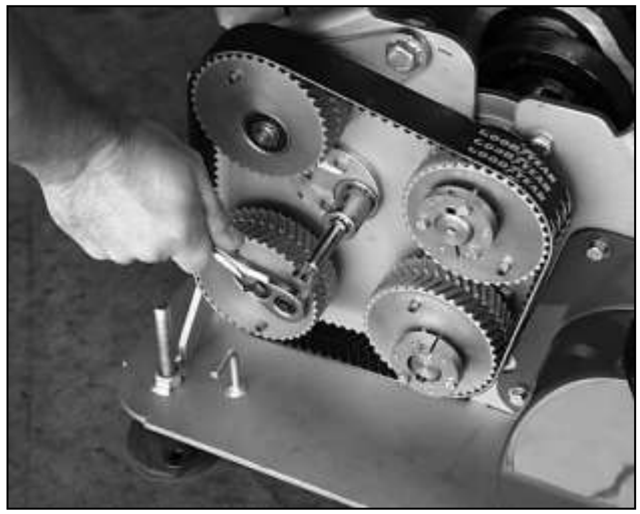

Photo 5-45 - Loosening the Tensioner Cam Pivot Bolt

7 **Tighten eccentric socket system**

Using a torque wrench, tighten the tensioner cam to 42 foot-pounds (see Photo 5-46). Tighten the tensioner cam pivot bolt to hold the sprocket position.

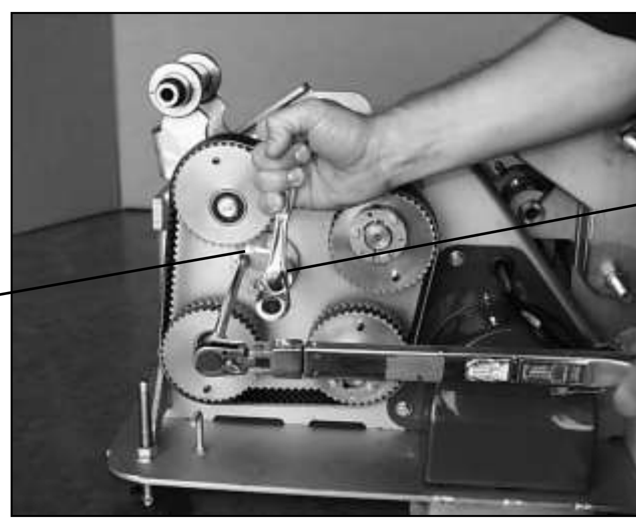

Photo 5-46 – Tightening the Tensioner Cam

Step **Description**

7 Tighten eccentric sprocket system (continued)

Tighten the jam nut on the back of the tensioner sprocket to 103 foot-pounds (see Photo 5-47).

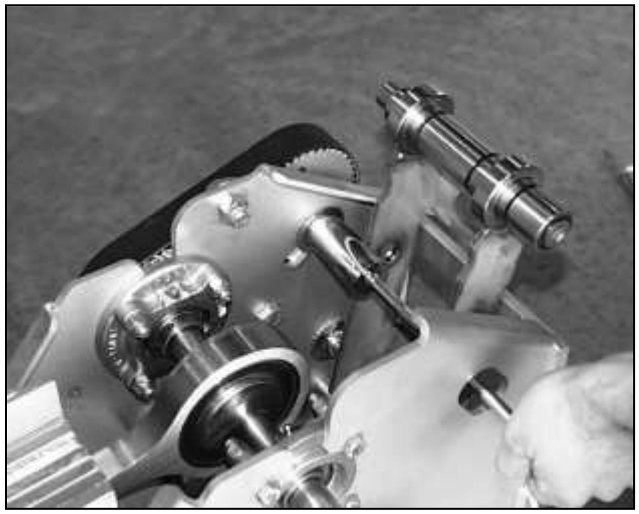

Photo 5-47 – Tightening the Tensioner Sprocket Bolt

8 Replace drive arm connecting rod

Replace the crankshaft connecting rod in the drive arm. Tighten the bolt to 42 foot-pounds (see Photo 5-48).

Photo 5-48 – Tightening the Drive Arm Connecting Rod

Step	**Description**

9 **Replace pan attachment arms**

Insert the bushing bolts for the pan attachment arms. Do not tighten the bolts yet.

10 **Orient idler arm in neutral position**

Rotate the crankshaft until the idler arm is perpendicular to the chassis base plate. Use a small machinist's square to check the idler arm position, as shown in Photo 5-49.

Photo 5-49 – Checking Position of Idler Arm

11 **Tighten drive arm pivot bolt**

Tighten the drive arm pivot bolt through the lower torsion bushing to 42 foot-pounds (see Photo 5-50).

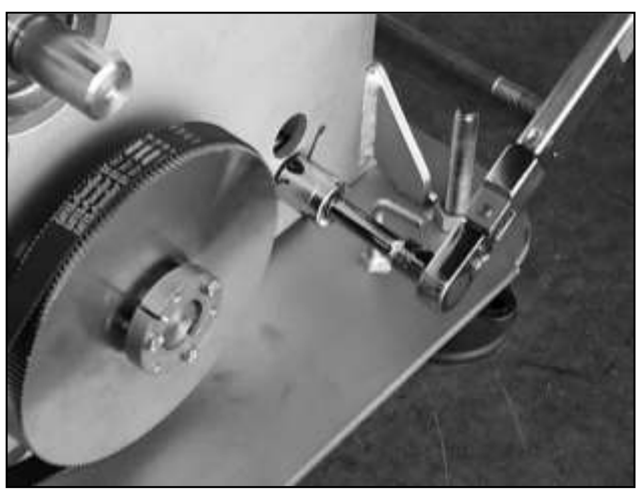

Photo 5-50 – Tightening the Drive Arm Pivot Bolt

Step	**Description**

12 Tighten pan attachment arm bolts

Tighten the pan attachment arm bolts to 42 foot-pounds, as shown in Photo 5-51.

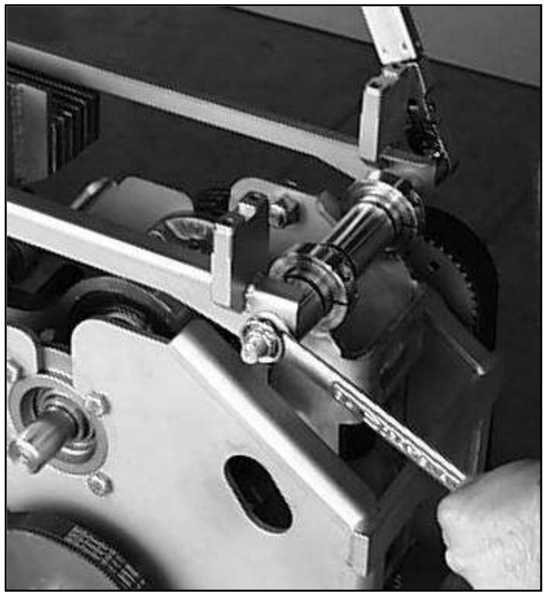

Photo 5-51 – Tightening the Pan Attachment Arm Bolts

13 Replace enclosure and pan assembly

Replace the drive assembly enclosure and pan assembly as described in “Replacing the Enclosure” and “Replacing the Pan Assembly.”

Replacing the Eccentric Timing Belt

This procedure describes how to replace the eccentric timing belt on the eccentric sprocket system.

Tools Needed: ¾-inch deep well socket and wrench, 3/4-inch socket and extension, torque wrench set 42 and 103 foot-pounds, timing plate (part no. 30547820), 15/16-inch socket, small machinist's square

Step	Description
1	**Remove pan and enclosure**
	Remove the pan and enclosure assemblies as described in "Removing the Pan Assembly" and "Removing the Enclosure."
2	**Loosen pan attachment arms**
	Using the ¾-inch wrench and socket, loosen the two bolts on each end of the pan attachment arms. Remove the bolt that goes through the drive arm, shown in Photo 5-52.

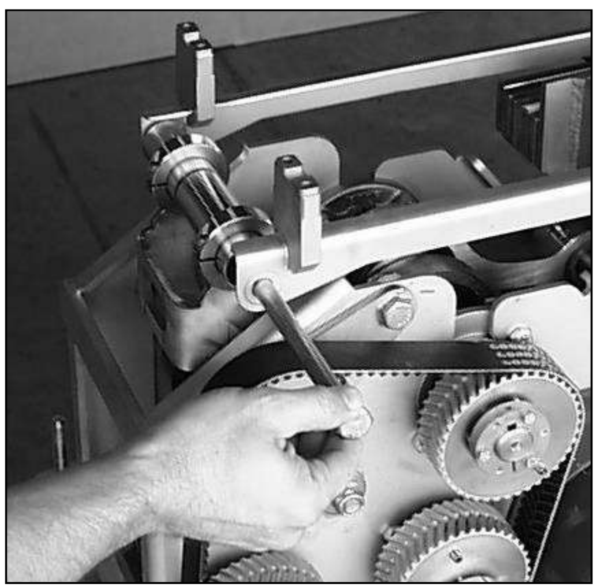

Photo 5-52 - Removing the Pan Attachment Arm Bolt

Step **Description**

3 **Disconnect drive arm from drive arm connecting rod**

Using a ¾-inch socket and extension through the chassis access hole, remove the bolt from the drive arm that holds the crank arm in place (see Photo 5-53).

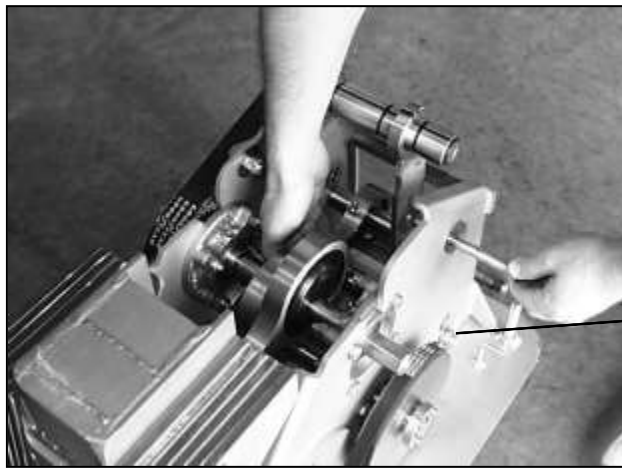

Photo 5-53 - Removing the Drive Arm Bolt

CAUTION ***Support the connecting rod with your free hand to prevent it from dropping and damaging the seals in the bearing.***

Using the ¾-inch socket and wrench, loosen the lower drive arm bushing bolt so that the drive arm swings freely. Lay the drive arm against the chassis.

Step	**Description**
4	**Loosen tensioner sprocket pivot nut**

Using the ¾-inch wrench, loosen the tensioner sprocket pivot nut one-quarter turn (see Photo 5-54). The nut may be on the opposite side of the frame from that shown.

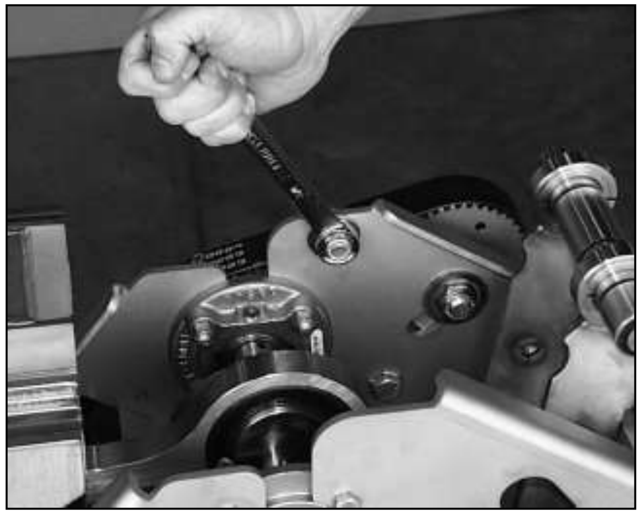

Photo 5-54 - Loosening the Tensioner Sprocket Pivot Nut

5	**Loosen jam nut behind tensioner sprocket stud**

Using the 15/16-inch socket, loosen the jam nut at the back of the tensioner sprocket stud by about one-half turn (see Photo 5-55).

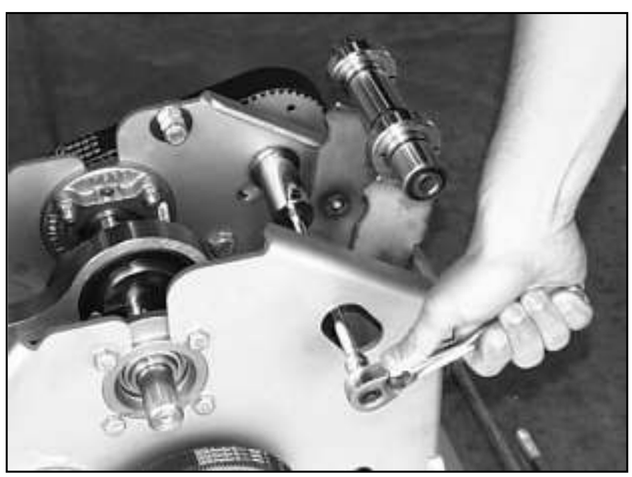

Photo 5-55 - Loosening the Tensioner Sprocket Stud

Step	Description

6 **Loosen nut on tensioner cam pivot bolt**

Using the ¾-inch socket, loosen the nut on the tensioner cam pivot bolt (see Photo 5-56).

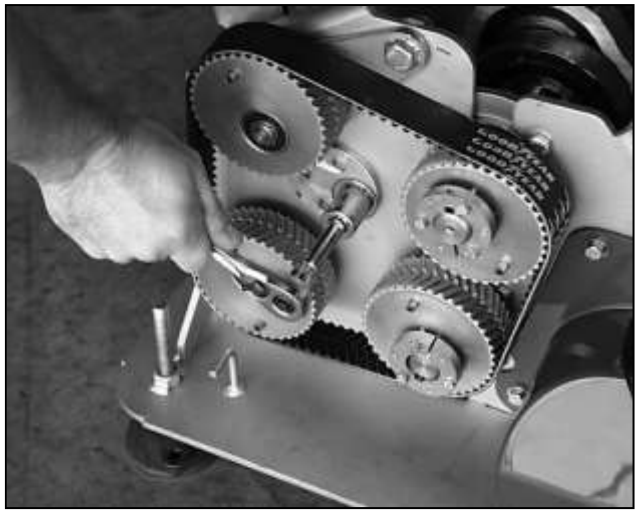

Photo 5-56 - Loosening the Tensioner Cam Pivot Bolt

Pivot the tensioner sprocket down to relieve the belt tension.

7 **Remove eccentric timing belt**

Remove the eccentric timing belt (see Photo 5-57).

Photo 5-57 – Removing the Eccentric Timing Belt

Step **Description**

8 Time eccentric sprockets

Time the crankshaft and driveshaft sprockets by placing the timing plate (part no. 30547820) over the flanges of their tapered bushings and aligning the pins in the slots of the plate (see Photo 5-58).

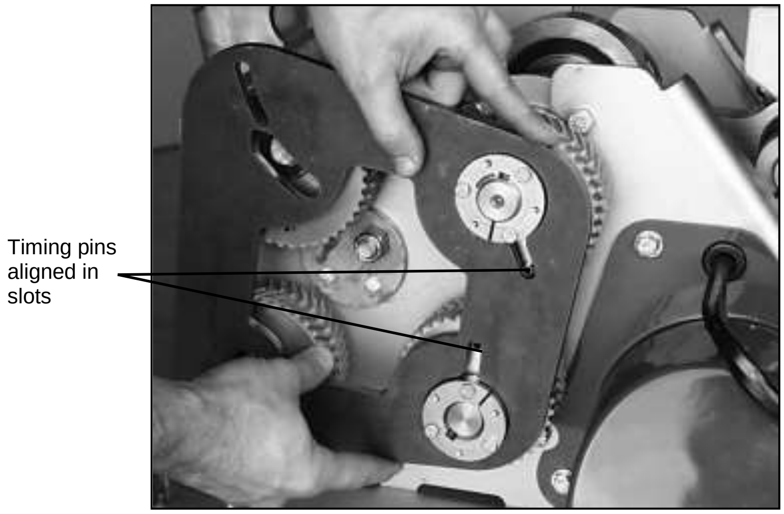

Photo 5-58 - Timing the Crankshaft and Driveshaft Sprockets

Time the tensioner and idler sprockets in the same way. Ensure that the tensioner sprocket is rotated downward in its slot and that the nut on the back side of the tensioner sprocket stud is not more than one-quarter turn loose. (see Photo 5-59).

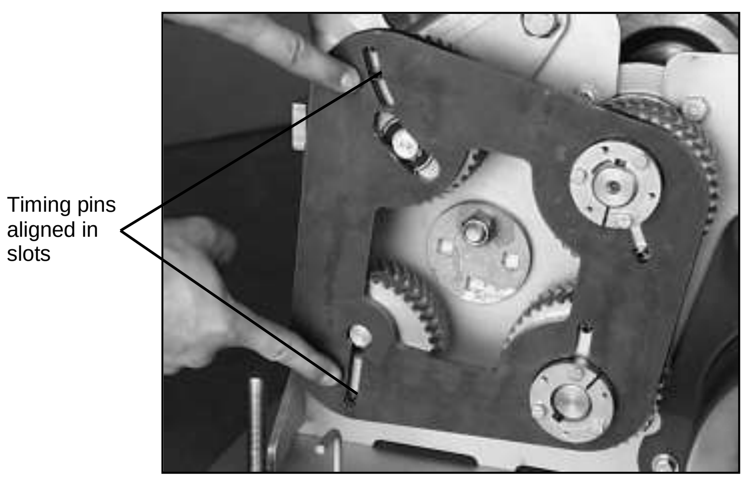

Photo 5-59 - Timing the Tensioner and Idler Sprockets

Step **Description**

9 **Install new eccentric timing belt**

Slip the new eccentric timing belt over all four sprockets as shown in Photo 5-60. Make sure there is no slack between the crankshaft, drive and idler sprockets.

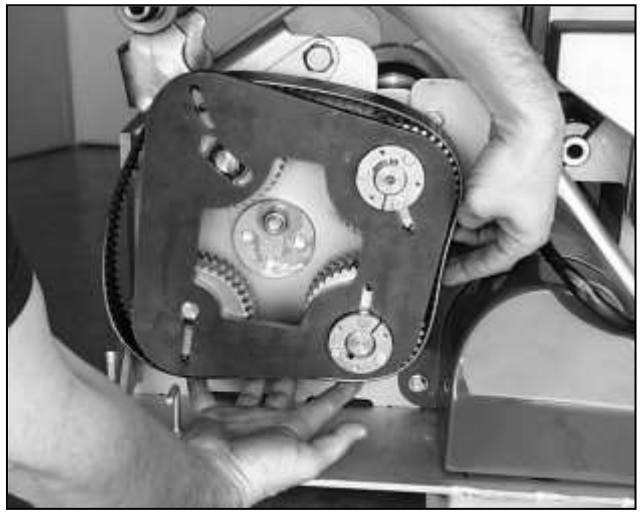

Photo 5-60 – Replacing the Eccentric Timing Belt

10 **Remove slack from eccentric timing belt**

With the timing plate still in place, apply a clockwise torque to the tensioner cam until all the slack is removed from the eccentric timing belt. The pin in the tensioner sprocket will be in the upper end of its slot in the timing plate as shown in Photo 5-61. Tighten the tensioner cam pivot bolt slightly to hold the tensioner in place and remove the timing plate.

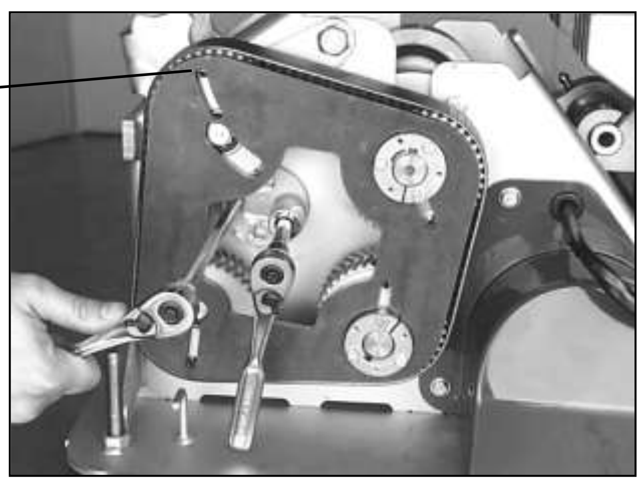

Photo 5-61 – Removing Slack from the Eccentric Timing Belt

Step **Description**

11 **Tighten eccentric sprocket system**

Tighten slightly (snug) the tensioner sprocket pivot nut (see Photo 5-62).

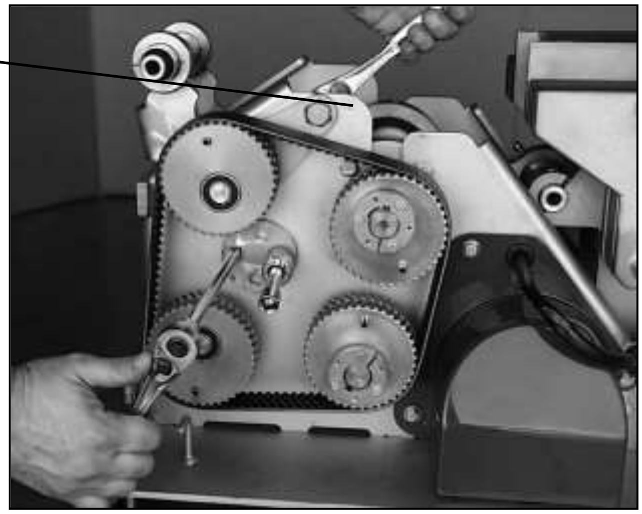

Photo 5-62 – Tightening the Tensioner Sprocket Pivot Nut

Using a torque wrench, tighten the tensioner cam to 42 foot-pounds (see Photo 5-63). Tighten the tensioner cam pivot bolt to 42 foot-pounds to hold the sprocket position.

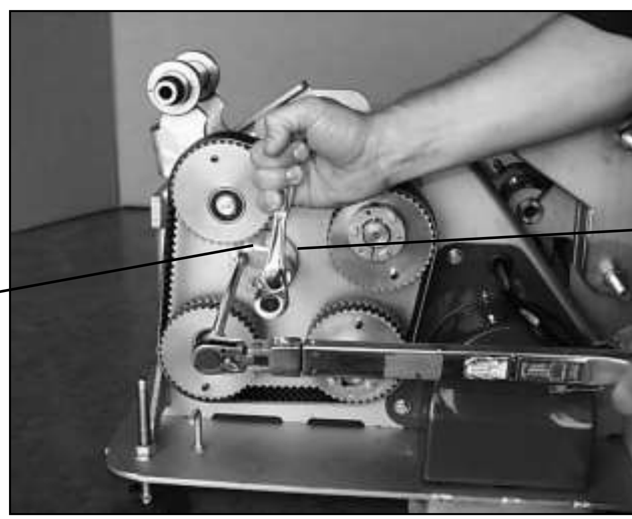

Photo 5-63 – Tightening the Tensioner Cam and Tensioner Cam Pivot Bolt

Step **Description**

11 **Tighten eccentric sprocket system (continued)**

Tighten the jam nut on the back of the tensioner sprocket stud to 103 foot-pounds (see Photo 5-64).

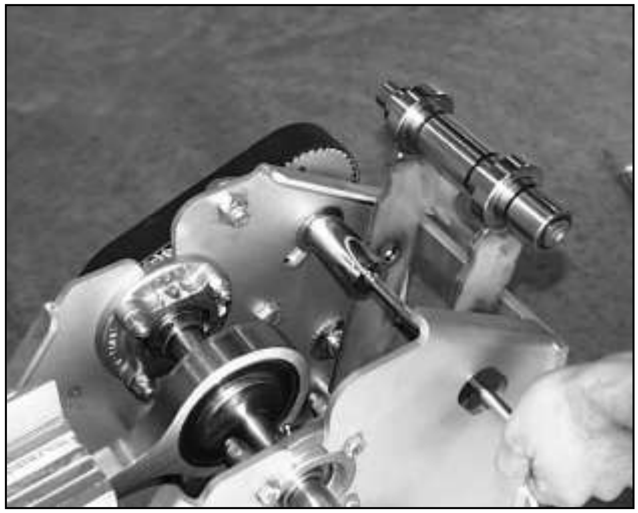

Photo 5-64 – Tightening the Tensioner Sprocket Jam Nut

12 **Replace drive arm connecting rod**

Replace the crankshaft connecting rod in the drive arm. Tighten the bolt to 42 foot-pounds (see Photo 5-65).

Photo 5-65 – Tightening the Drive Arm Connecting Rod

Step **Description**

13 Replace pan attachment arms

Insert the bushing bolts for the pan attachment arms. Do not tighten the bolts yet.

14 Orient idler arm in neutral position

Rotate the crankshaft until the idler arm is perpendicular to the chassis base plate. Use a small machinist's square to check the idler arm position, as shown in Photo 5-66.

Photo 5-66 – Checking Position of Idler Arm

15 Tighten drive arm pivot bolt

Tighten the drive arm pivot bolt through the lower torsion bushing to 42 foot-pounds (see Photo 5-67).

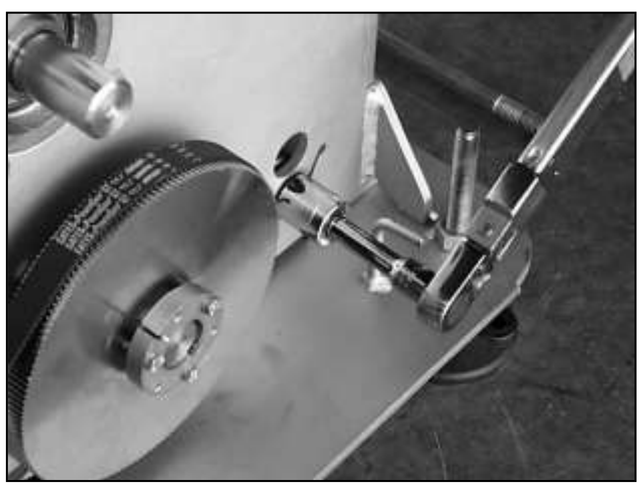

Photo 5-67 – Tightening the Drive Arm Pivot Bolt

Step **Description**

16 Tighten pan attachment arm bolts

Tighten the pan attachment arm bolts to 42 foot-pounds, as shown in Photo 5-68.

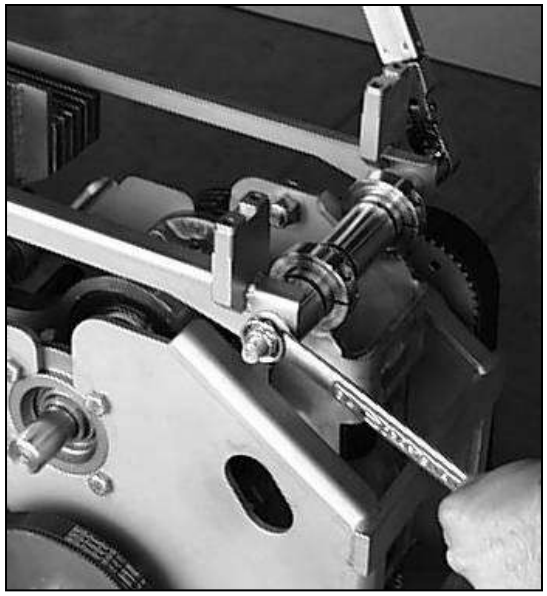

Photo 5-68 – Tightening the Pan Attachment Arm Bolts

17 Replace enclosure and pan assembly

Replace the drive assembly enclosure and pan assembly as described in “Replacing the Enclosure” and “Replacing the Pan Assembly.”

Replacing the Crankshaft Assembly

This procedure describes how to replace the crankshaft assembly. The replacement assembly consists of a crankshaft with pre-lubricated bearings and two connecting rods. The flange bearings and crankshaft sprocket are reused.

Tools Needed: ¾-inch deep well socket and wrench, 9/16-inch socket and wrench, 1/8-inch,3/16-inch, and 3/32-inch allen wrenches, torque wrench that can be set to 54, 65 and 110 inch-pounds, torque wrench that can be set to 20, 42 and 103 foot-pounds, deadblow hammer, 5/16-inch socket, 12-inch straightedge, timing plate (part no. 30547820), two 3/8-inch ratchet handles with at least a 4-inch extension, 15/16-inch socket, extensions of various lengths, small machinist's square

Step **Description**

1 Remove pan and enclosure

Remove the pan and enclosure assemblies as described in "Removing the Pan Assembly" and "Removing the Enclosure."

2 Move pan attachment arms out of the way

Using the ¾-inch wrench and socket, remove the bolt from the pan attachment arms above the eccentric sprocket system as shown in Photo 5-69.

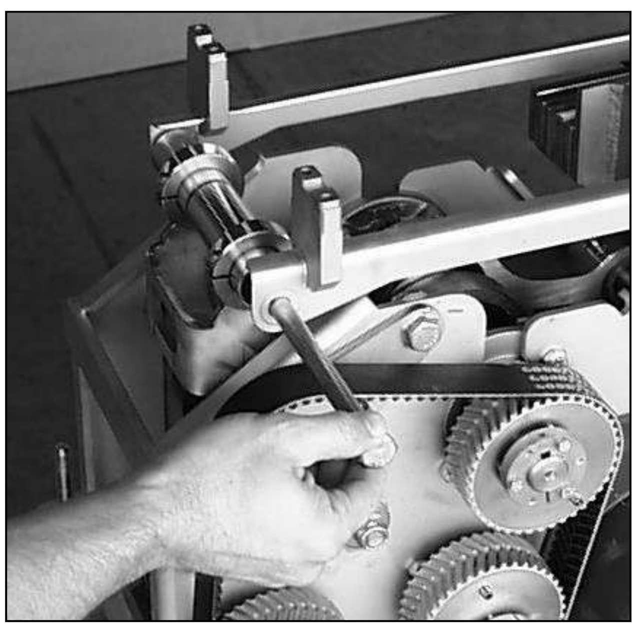

Photo 5-69 - Removing the Pan Attachment Arm Bolt

Step **Description**

2 **Move pan attachment arms out of the way (continued)**

Using the ¾-inch wrench and socket, loosen the pan attachment arm bolt at the counterweight end. Swing the pan attachment arms out of the way (see Photo 5-70).

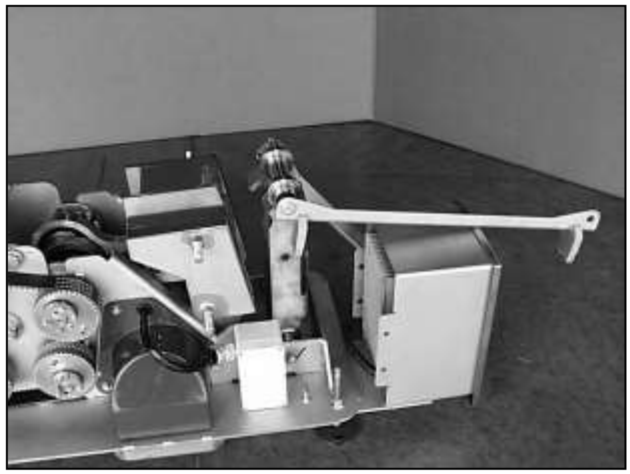

Photo 5-70 - Pan Attachment Arms Moved Away

3 **Disconnect drive arm from drive arm connecting rod**

Using a ¾-inch socket and extension through the chassis access hole, remove the bolt from the drive arm that holds the crank arm in place (see Photo 5-71).

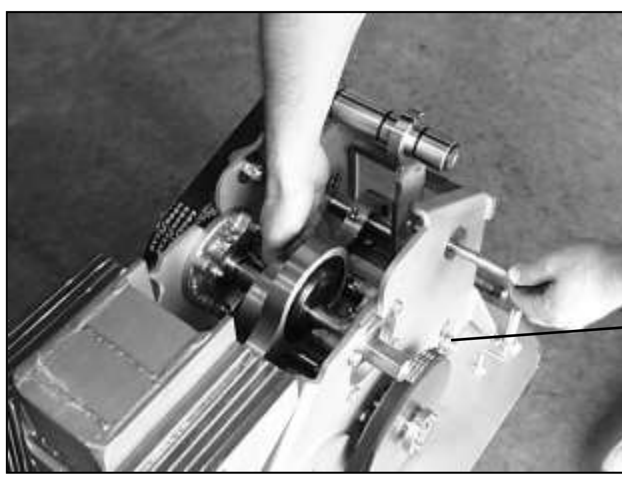

Photo 5-71 - Removing the Drive Arm Bolt

Step	**Description**

3 **Disconnect drive arm from drive arm connecting rod (continued)**

CAUTION

Support the connecting rod with your free hand to prevent it from dropping and damaging the seals in the bearing.

Using the ¾-inch socket and wrench, loosen the lower drive arm bushing bolt so that the drive arm swings freely. Lay the drive arm against the chassis.

4 **Loosen tensioner sprocket pivot nut**

Using the ¾-inch wrench, loosen the tensioner sprocket pivot nut one-quarter turn (see Photo 5-71). The nut may be on the opposite side of the frame from that shown.

Photo 5-71 - Loosening the Tensioner Sprocket Pivot Nut

Step **Description**

5 Loosen jam nut behind tensioner sprocket stud

Using the 15/16-inch socket, loosen the jam nut at the back of the tensioner sprocket stud about one-half turn (see Photo 5-72).

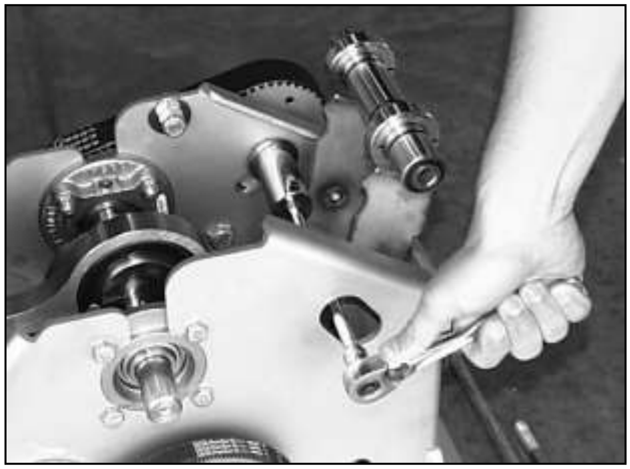

Photo 5-72 - Loosening the Tensioner Sprocket Stud

6 Loosen nut on tensioner cam pivot bolt

Using the ¾-inch socket, loosen the nut on the tensioner cam pivot bolt (see Photo 5-73).

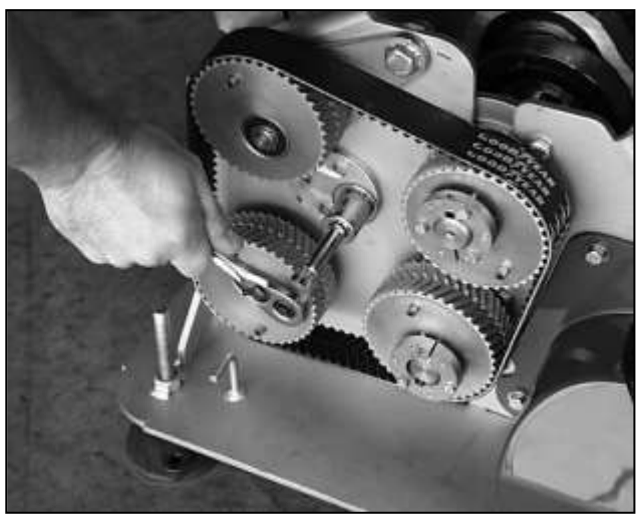

Photo 5-73 - Loosening the Tensioner Cam Pivot Bolt

Pivot the tensioner sprocket down to relieve the belt tension.

Step **Description**

7 **Remove eccentric timing belt**

Remove the eccentric timing belt (see Photo 5-74).

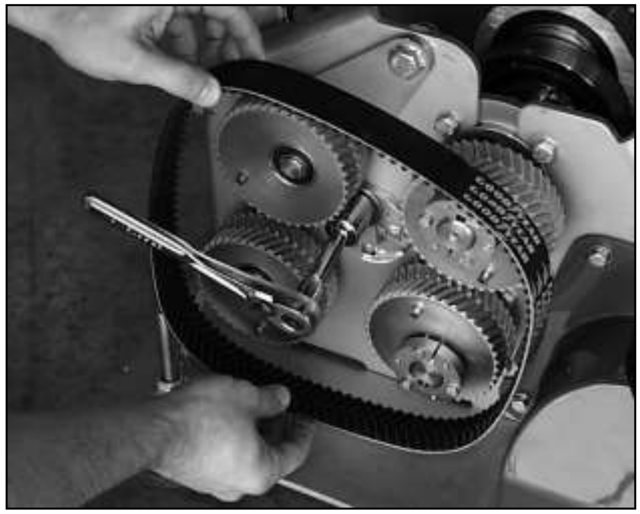

Photo 5-74 – Removing the Eccentric Timing Belt

8 **Disconnect counterweight from counterweight connecting rod**

Using the 3/16-inch allen wrench, remove the screws on the counterweight split collars (see Photo 5-75).

CAUTION

Support the connecting rod with your free hand to prevent it from dropping and damaging the seals in the bearing.

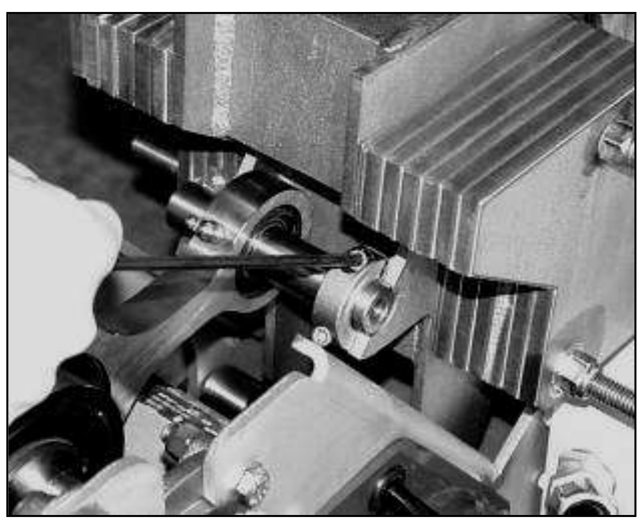

Photo 5-75 - Removing the Split Collar Screws

Step	**Description**
8	**Disconnect counterweight from counterweight connecting rod (continued)**
	Remove the counterweight connecting rod from the counterweight split collars.
9	**Remove old crankshaft assembly**
	Using a 9/16-inch socket and wrench, remove the four bolts from each crankshaft flange bearing. Rotate the crankshaft sprocket so that each bolt head clears the sprocket (see Photo 5-76).

Photo 5-76 - Removing Bolts from Flange Bearings

Using a 1/8-inch allen wrench, loosen the set screws on the crankshaft flange bearings, as shown in Photo 5-77.

Photo 5-77 – Loosening the Flange Bearing Set Screws

Step **Description**

9 Remove old crankshaft assembly (continued)

Push the flange bearing closest to the timing sprocket at least ¾ inch toward the eccentric lobes on the crankshaft. Be careful not to damage the crankshaft journals. Then push the opposite flange bearing with the crankshaft assembly out of the chassis side plate (see Photo 5-78).

Note: Photo 5-78 shows the tensioner sprocket pivot nut removed so that it does not interfere with the bearing flange.

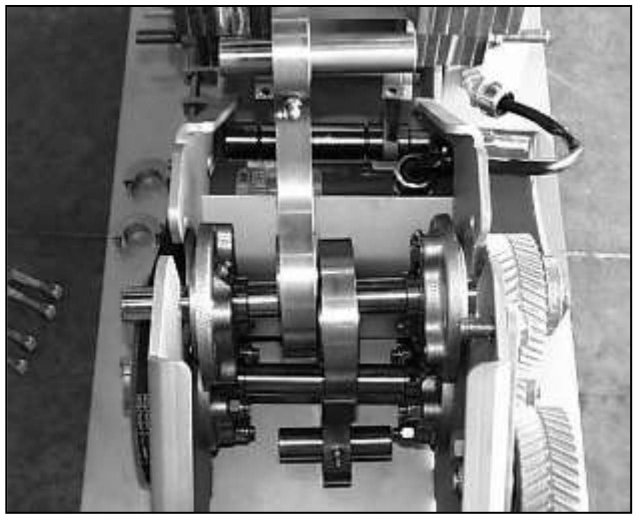

Photo 5-78 – Pushing Flange Bearings toward Center of Drive

Lift the crank assembly out of the conveyor (see Photo 5-79).

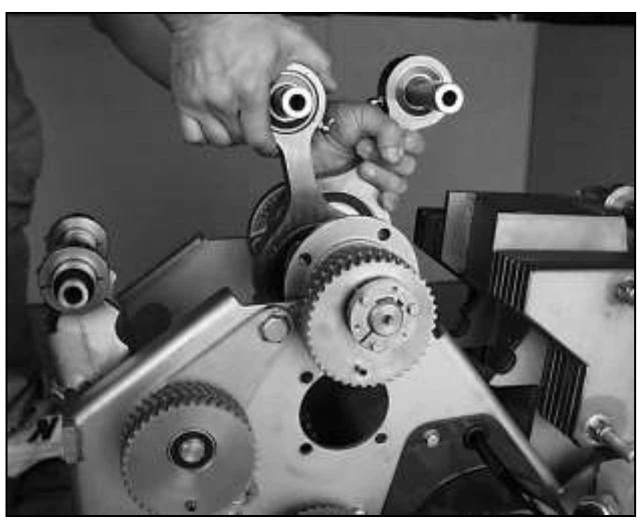

Photo 5-79 – Removing the Crankshaft Assembly

Step **Description**

9 **Remove old crankshaft assembly (continued)**

Transfer the flange bearings and sprocket from the old crankshaft assembly to the new one. Place the key for the sprocket bushing in the corresponding slot on the new crankshaft.

CAUTION

Product travel direction will be reversed if the sprocket bushing key is installed incorrectly.

Do not tighten the bushing screws or bearing set screws at this time.

10 **Install new crankshaft assembly**

Install the new crankshaft assembly so that the flange bearings are flush with the chassis. Use at least two bolts to hold each bearing against the frame.

Replace the tensioner sprocket pivot nut if it was removed. Tighten it until all play is removed.

With a deadblow hammer (see Photo 5-80), gently tap the crankshaft assembly into position so that the drive arm connecting rod bushing is centered and slides freely up and down in the drive arm (see Photo 5-81).

***Note: Make sure the bearing set screws are loose*.**

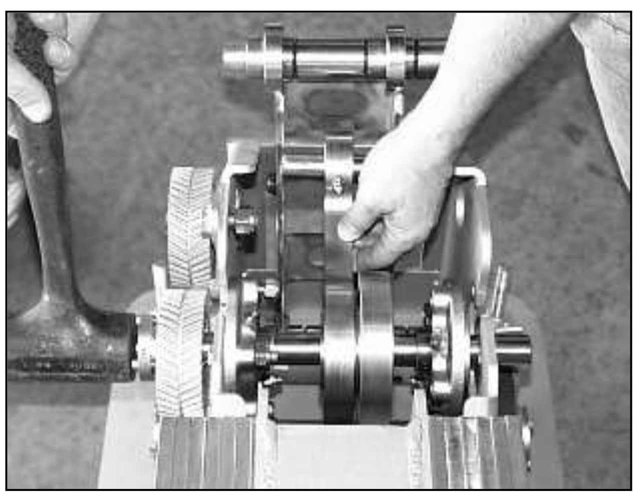

Photo 5-80 – Positioning the Crankshaft Assembly

Step **Description**

10 **Install new crankshaft assembly (continued)**

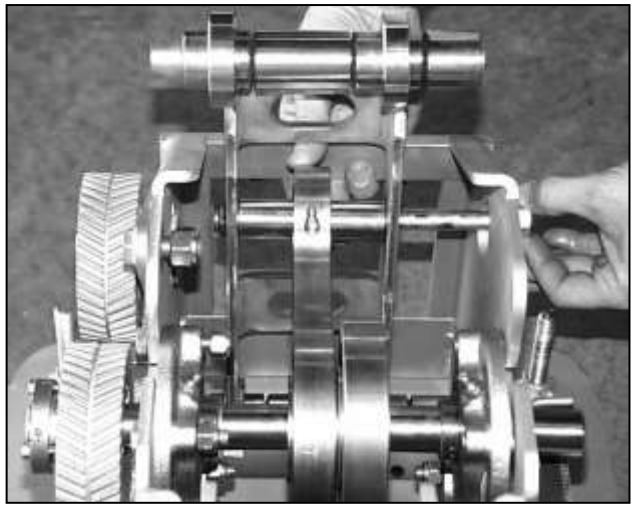

Photo 5-81 – Centering the Bushing in the Drive Arm

Insert the drive arm connecting rod bolt to keep the connecting rod from falling.

11 **Tighten bearing mounting bolts**

Using a torque wrench set to 20 foot-pounds and a 9/16-inch socket and wrench, tighten the eight flange bearing mounting bolts onto the chassis (see Photo 5-82).

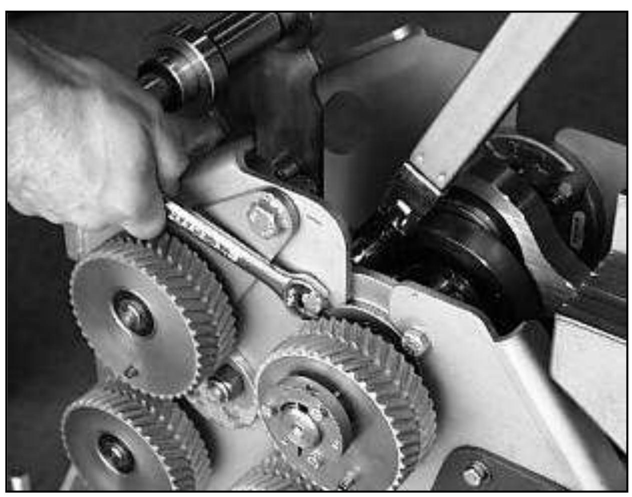

Photo 5-82 – Tightening Bearing Mounting Bolts

<u>Step</u> **<u>Description</u>**

12 Tighten set screws

Using a torque wrench set to 65 in-lbs. and a 1/8-inch allen wrench that is not worn, tighten the bearing set screws onto the crankshaft (see Photo 5-83).

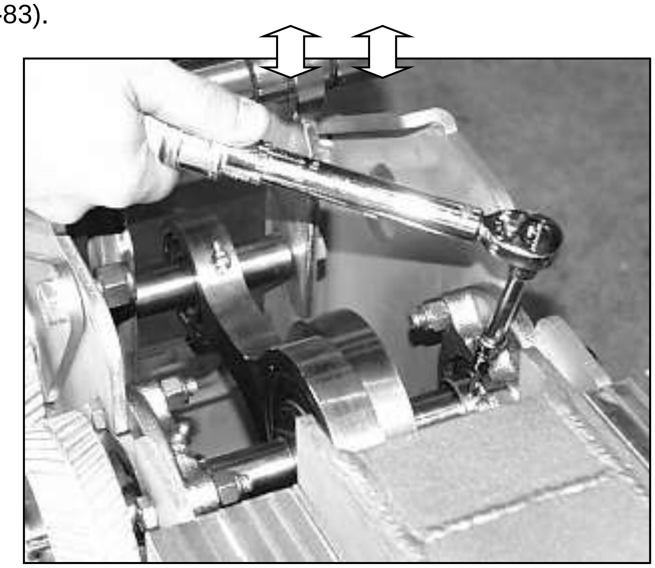

Photo 5-83 – Tightening Bearing Set Screws

13 Install crankshaft arm into counterweights

Using a torque wrench set to 110 inch-pounds and a 3/16-inch allen wrench, install the counterweight connecting rod bushing into the split collars on the counterweight assembly.

Step	**Description**
14	**Realign eccentric sprockets**

Check the position of the mounted sprocket on the new crankshaft by placing a straightedge against the face of the drive sprocket (see Photo 5-84). Both sprockets should be flush with the straightedge within 1/64 inch when the bushings are fully tightened. If they are not, the eccentric timing belt may fail prematurely.

Both sprockets flush with straightedge

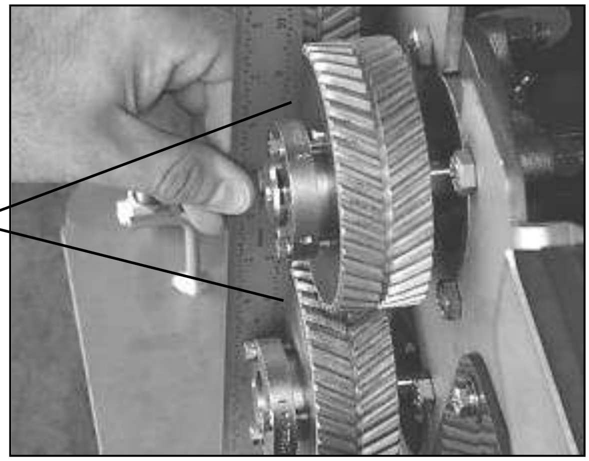

Photo 5-84 – Checking the Position of the Mounted Sprocket

If the crankshaft sprocket is not flush with the other three sprockets, loosen the bushing and reposition the sprocket along the crankshaft. The tapered bushings will typically draw a sprocket up onto them 3/32 inch. Take this into consideration when initially positioning the sprocket and bushing before tightening. Torque the tapered bushing bolts to 54 inch-pounds as shown in Photo 5-85. Tighten the bushing set screws using the 3/32-inch allen wrench.

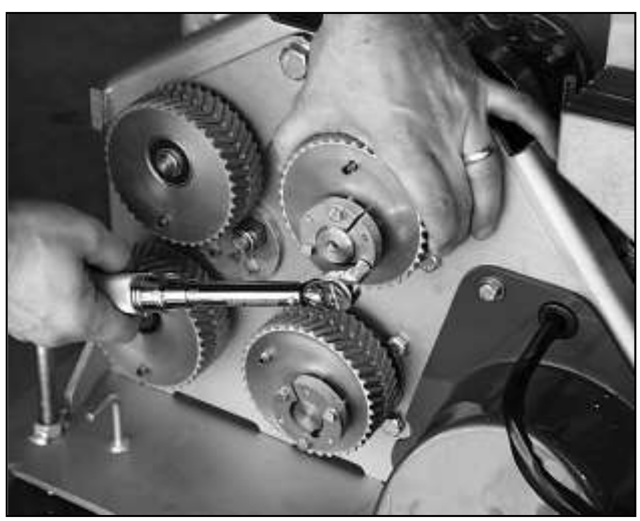

Photo 5-85 – Tightening the Bushing Bolts

Step	**Description**
15	**Time the eccentric sprockets**

Remove the bolt from the drive arm connecting rod to disconnect the drive arm from the crank assembly and lay it against the chassis.

Time the crankshaft and driveshaft sprockets by placing the timing plate (part no. 30547820) over the flanges of their tapered bushings and aligning the pins in the slots of the plate (see Photo 5-86).

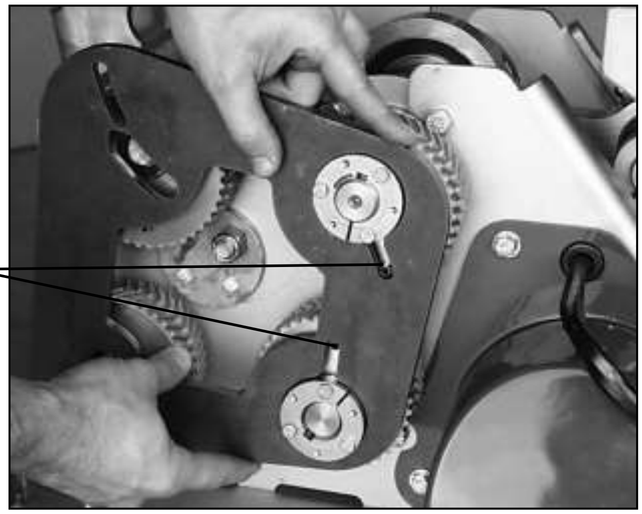

Photo 5-86 - Timing the Crankshaft and Driveshaft Sprockets

Time the tensioner and idler sprockets in the same way. Ensure that the tensioner sprocket is rotated downward in its slot and that the nut on the back side of the tensioner sprocket stud is not more than one-half turn loose. (see Photo 5-87).

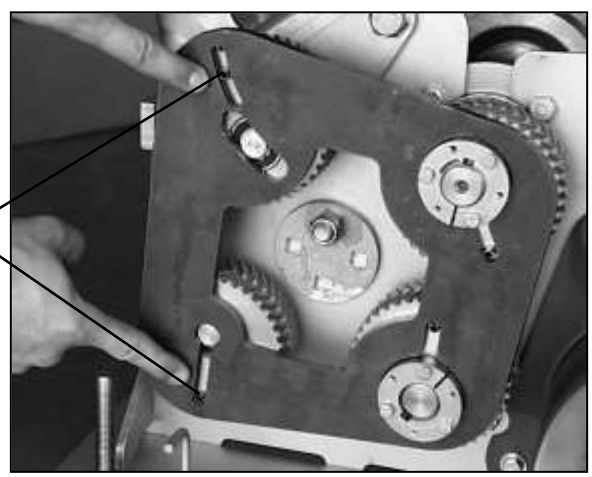

Photo 5-87 - Timing the Tensioner and Idler Sprockets

Step **Description**

16 **Replace eccentric timing belt**

Slip the eccentric timing belt over all four sprockets as shown in Photo 5-88. Make sure there is no slack between the crankshaft, drive and idler sprockets.

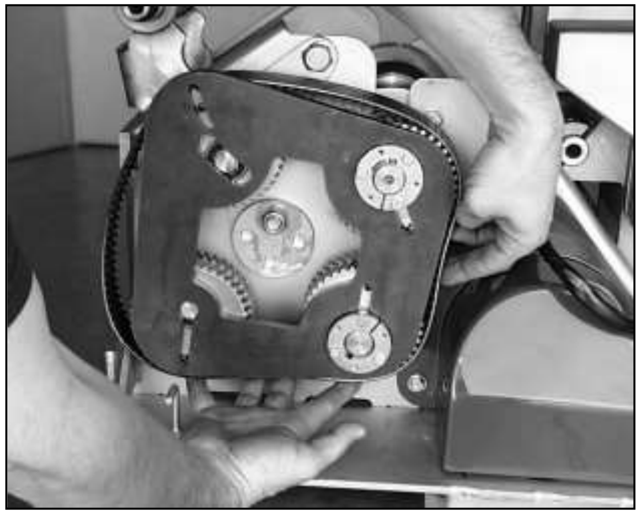

Photo 5-88 – Replacing the Eccentric Timing Belt

17 **Remove slack from eccentric timing belt**

With the timing fixture still in place, apply a clockwise torque to the tensioner cam until all the slack is removed from the eccentric timing belt. The pin in the tensioner sprocket will be in the upper end of its slot in the timing fixture as shown in Photo 5-89. Tighten the tensioner cam pivot bolt slightly to hold the tensioner in place and remove the timing plate.

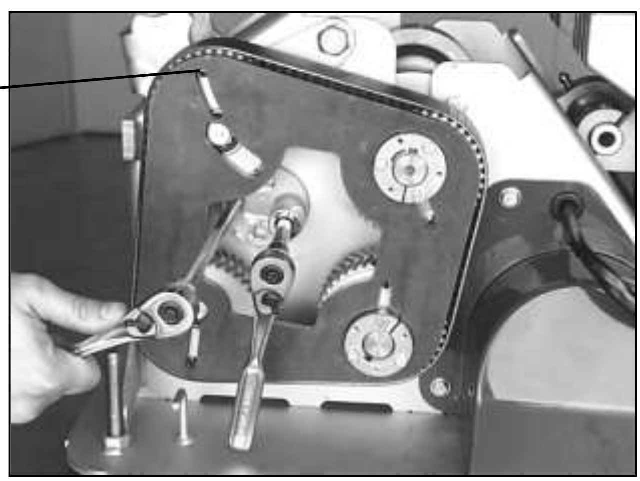

Photo 5-89 – Removing Slack from the Eccentric Timing Belt

Step	**Description**
18	**Tighten eccentric sprocket system**

Tighten slightly (snug) the tensioner sprocket pivot nut (see Photo 5-90).

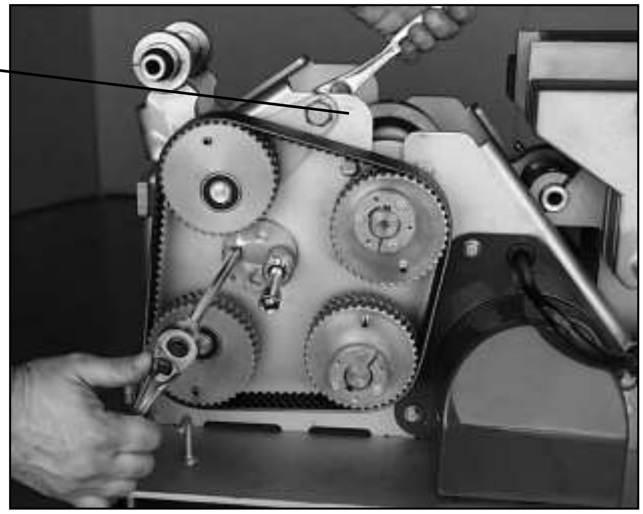

Photo 5-90 – Tightening the Tensioner Sprocket Pivot Nut

Using a torque wrench, tighten the tensioner cam to 42 foot-pounds (see Photo 5-91). Tighten the tensioner cam pivot bolt to 42-foot-pounds to hold the sprocket position.

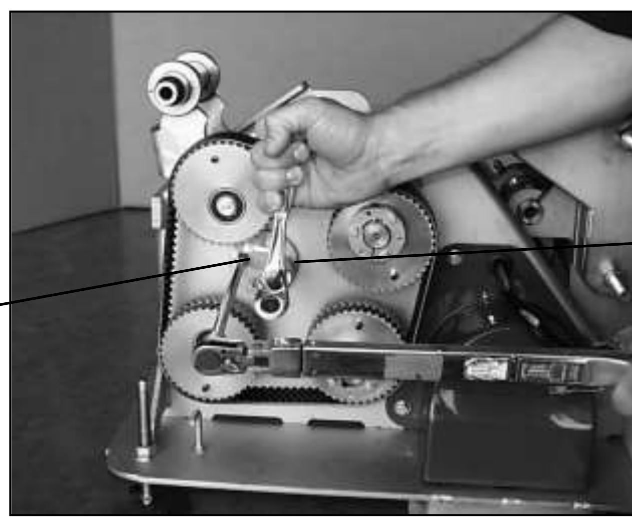

Photo 5-91 – Tightening the Tensioner Cam and Tensioner Cam Pivot Bolt

Step **Description**

18 Tighten eccentric sprocket system (continued)

Tighten the jam nut on the back of the tensioner sprocket stud to 103 foot-pounds (see Photo 5-92).

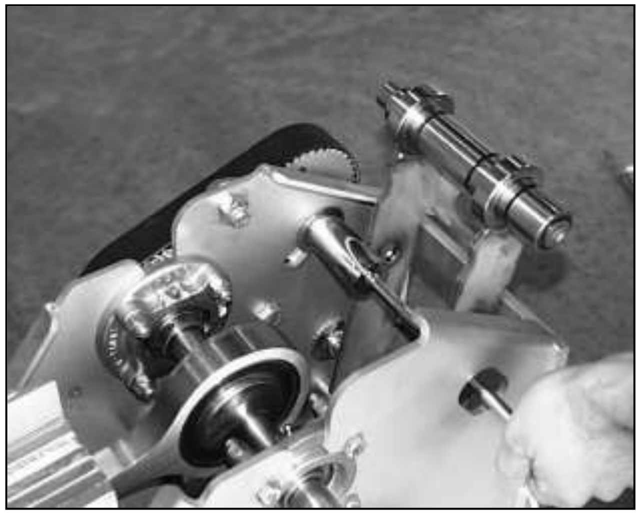

Photo 5-92 – Tightening the Tensioner Sprocket Jam Nut

19 Replace drive arm connecting rod

Replace the crankshaft connecting rod in the drive arm. Tighten the bolt to 42 foot-pounds (see Photo 5-93).

Photo 5-93 – Tightening the Drive Arm Connecting Rod

Step	**Description**

20 **Replace pan attachment arms**

Insert the bushing bolts for the pan attachment arms. Do not tighten the bolts yet.

21 **Orient idler arm in neutral position**

Rotate the crankshaft until the idler arm is perpendicular to the chassis base plate. Use a small machinist's square to check the idler arm position, as shown in Photo 5-94.

Photo 5-94 – Checking Position of Idler Arm

22 **Tighten drive arm pivot bolt**

Tighten the drive arm pivot bolt through the lower torsion bushing to 42 foot-pounds (see Photo 5-95).

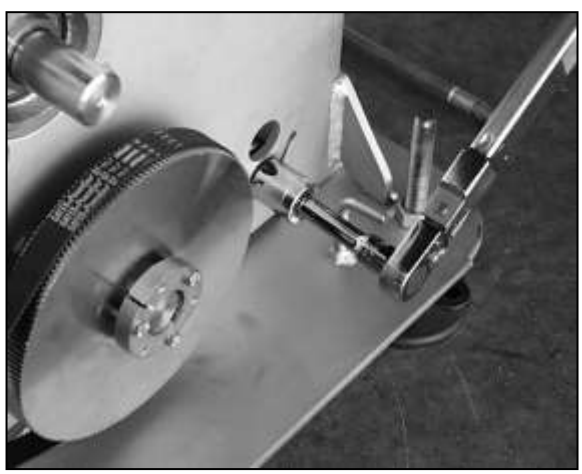

Photo 5-95 – Tightening the Drive Arm Pivot Bolt

Step	**Description**
23	**Tighten pan attachment arm bolts**

Tighten the pan attachment arm bolts to 42 foot-pounds, as shown in Photo 5-96.

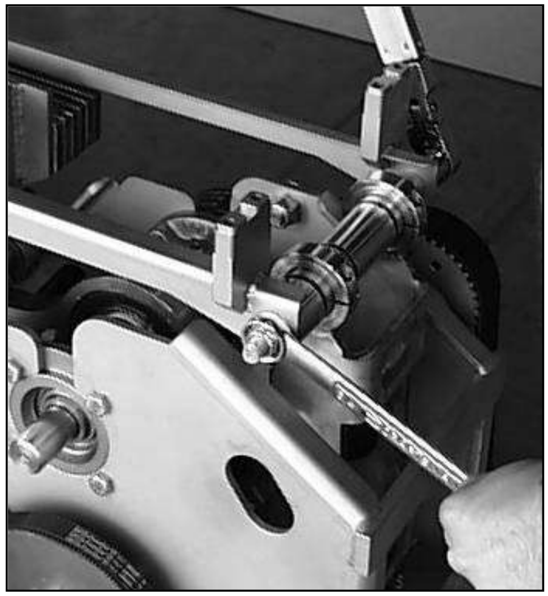

Photo 5-96 – Tightening the Pan Attachment Arm Bolts

24 Replace enclosure and pan assembly

Replace the drive assembly enclosure and pan assembly as described in “Replacing the Enclosure” and “Replacing the Pan Assembly.”